Forecasting Retail Sales at Every Grain: Trees, Foundation Models, Transformers and World Models Compared on Four Public Data Sets

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Abstract. A retailer makes decisions from the same demand at different grains and on different cycles: store orders by item and day, promotions by item and week in blocks of weeks, category reviews week by week, and supplier ranges and terms over a year. It wants one forecast that serves all of them without losing value between plans. This paper asks whether one forecast, built at the finest grain and summed to every coarser grain, can serve these decisions across every kind of line, and which kind of model does that best. It compares gradient-boosted trees (LightGBM), pretrained foundation models (TimesFM, Chronos-Bolt, Timestral), an attention-based model (the Temporal Fusion Transformer) and a world model that carries each item's state forward under the price and promotion plan, on four public data sets (M5, six Dominick's categories, Breakfast at the Frat and Corporación Favorita), with the world model and LightGBM tuned by a common search procedure. Results are read by grain, by the nature of the line, on seen and unseen data, and across retailers. On lines and stores the models were trained on, the tuned world model and LightGBM are close, each ahead on different data sets and grains, and their average is the most accurate choice across grains: it has the lowest error in 22 of 27 grain and data set cases, and in paired item tests it is clearly more accurate than the world model in 109 of 240 comparisons (clearly less in 22) and than LightGBM in 106 (clearly less in 18). It is also the most accurate on the fast lines that carry most sales and on the slow tail on eight of the nine data sets. On items kept out of training, the world model loses at most 0.040 in scaled error and forecasts within 11% of actual sales, while LightGBM loses up to 0.311 and forecasts as little as two thirds of actual sales, which pulls the average below the world model alone on all five data sets tested. In the one transfer between retailers tested, settings tuned on M5 carried to Favorita, and training on other retailers' data improved neither model by more than 0.003. On these data, the rule that follows is one forecast for every item in every store, the average of the tuned world model and tuned trees summed to every grain, with the world model alone for items the models have not been trained on. The rule holds for the data sets studied; anyone adopting the framework should rerun every comparison, including those that lost here, on their own data. The study covers forecasts of 8 weeks and 28 days and the price and promotion levers; whether the models hold over 26- and 52-week plans is left for work with a retailer's own data.

Keywords: retail forecasting; forecast grain; hierarchical evaluation; forecast combination; promotions; world models; foundation models.

1 Introduction

Retail decisions run at different grains and on different clocks. Supplier range, space and cost reviews look at an item's volume across the chain over 26 or 52 weeks, and the terms agreed in them commit allowances for the year ahead. Promotions are planned item by week in blocks of six to twelve weeks, and the promotional and retail allowances that fund them follow the same blocks. Merchants judge their category week by week. Store inventory is ordered item by store, day by day [5]. Each of these decisions reads the same demand at a different grain and over a different span of time.

Since there is no single grain, there is no reason to expect one model to be the most accurate at every grain and every horizon. Retailers do not want a separate model at each grain either. Separate models disagree, and value leaks at each hand-off from a coarse long-range plan to a finer short-range one. The annual plan agreed with a supplier is broken into promotion blocks, weeks and store orders. Where the finer forecasts do not add back to the plan, the gap is paid for in stock, markdowns or lost sales. The forecasting literature treats this as coherence across the levels of the product and store hierarchy [30, 31] and across levels of time [26], usually by forecasting each level and reconciling the forecasts. This paper takes the other route retailers use: one forecast made at the finest grain, item by store by period, and summed to every coarser grain, so that the grains add up by construction. The question this paper answers is whether one forecast, built once at the finest grain, can serve the decisions a retailer makes from it, across every kind of

line, and which kind of model, or combination of models, does that best.

Lines differ as much as grains do. In grocery a short head of fast lines carries most of the units, and a long tail of slow lines sells a few units a week or none. On the Dominick's categories studied here, 11 to 21% of the items make 80% of unit sales. Ranges also change: items are added and delisted and stores open, so part of every forecast is for a line or a store with little or no history. A forecast that serves the fast lines well can fail on the tail or on a new item, and a planner deciding for one kind of line needs to know which.

We test four kinds of model. Gradient-boosted trees are the proven method: LightGBM was behind the leading entries of the M5 competition [1]. Pretrained foundation models (TimesFM, Chronos-Bolt and Timestral) forecast without training on the retailer's data. The Temporal Fusion Transformer is an attention-based model trained on it. World models, the newest of the four, learn a state for each item and carry it forward one period at a time under the price and promotion plan; ours is shown in Figure 2. The world model and LightGBM were each tuned by the common search procedure on every data set, and the average of forecasts is tested alongside them. The data are nine data sets from four retailers' public data: M5, six categories of Dominick's, Breakfast at the Frat, and Corporación Favorita. Six of the nine are categories of one retailer, so the evidence across retailers rests on four.

The study covers the short end of the planning cycle. Forecasts run 8 weeks ahead on the weekly data and 28 days ahead on the daily data, and are scored at every grain from item by store to the chain total, period by period and summed over the whole window as a promotion block. Forecasts of 26 or 52 weeks are not tested: judging them needs a retailer's own plans and range decisions to score against, which no public data set carries, and is left for work with an industry partner. Of the four Ps of retail, the study covers price and promotion, through the plan inputs these data record; product and place are out of scope. Every analysis was written down before it ran; the few added after the first results were read are listed in Section 17.

Section 2 sets out the framework, and the results follow it part by part: grain, the nature of the line, seen and unseen data, and transferability, with the broad view last.

Section 3 places the work among retail promotion models, hierarchical evaluation, forecast combination, pretrained forecasters and world models. Sections 4 to 6 give the data, the models and the experimental setup. Section 8 reports the results, Section 15 what was tried and dropped, and Sections 16 to 18 discuss what the results mean for a retailer, their limits, and conclude.

2 Framework

The overall question of this paper is:

> Can one forecast, built once at the finest grain of item by store by period and summed to every coarser grain, serve the decisions a retailer makes from it across every kind of line, and which kind of model, or combination of models, does that best?

It is answered through four research questions, in the order a retailer meets them:

RQ1, grain. At which grains of product, store and time does each model forecast best, period by period and over a whole promotion block, when every grain is summed from the same item-by-store forecast?

RQ2, nature of the line. How does each model do on the fast lines that carry most of the sales, on the long tail of slow lines, and on lines with short history?

RQ3, seen and unseen data. How much accuracy does each model lose on items and stores it was not trained on?

RQ4, transferability. Do a model's settings carry to another retailer, and does training on other retailers' data make it more accurate?

These are the questions retailers ask of a forecast in practice (Section 22). Each question fixes what is measured before any model is compared, and the results are reported in the same order (Section 8). The broad view then brings the four answers together.

RQ1, grain. Does the forecast hold up at every grain a decision is made at, and do the grains add up? Every model forecasts each item in each store for each period, and the forecasts are summed to item, store, category, category by store and the chain total. Each grain is scored period by period, which is how store orders and the category week are judged, and over the whole forecast window as one block, which is how a promotion is judged. Because every grain is a sum of the same item-by-store forecast, the plans built from them add up, and any value lost between grains shows as a difference in accuracy between them.

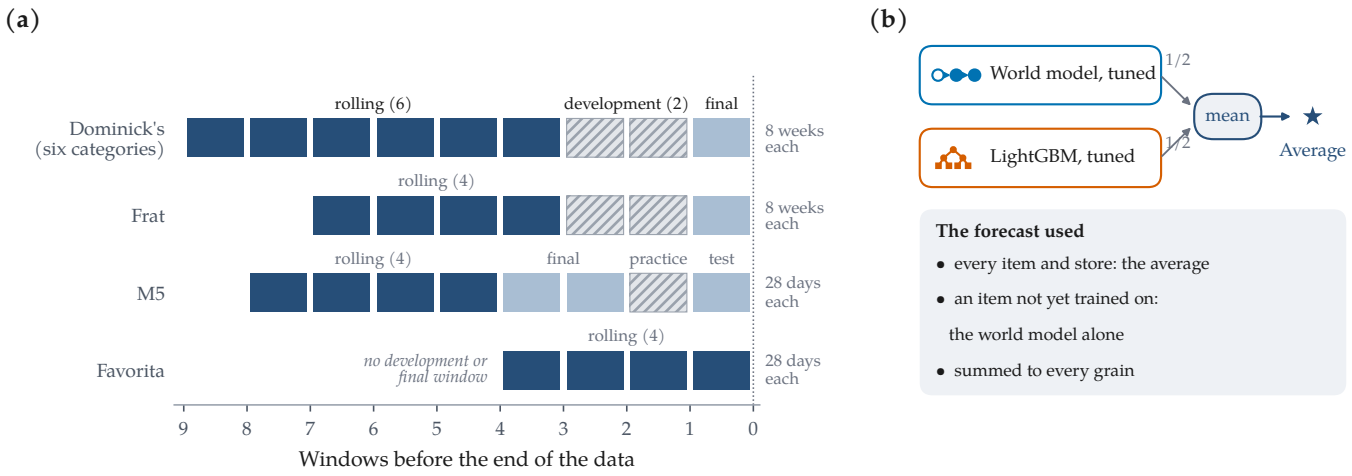


Figure 1: Every result of Section 8 is scored on rolling windows that no choice of setting saw, and the forecast recommended is the average of the two tuned models. (**a**) Windows by data set, in windows before the end of the data: the rolling windows scored in this paper; and the development and final windows of the first version, which are not scored here. The searches that tuned each model scored windows earlier than the rolling windows. (**b**) The forecast that the results support: the equal-weight average of the tuned world model and the tuned LightGBM for every item and store, the world model alone for an item the models have not yet been trained on, summed to every grain.

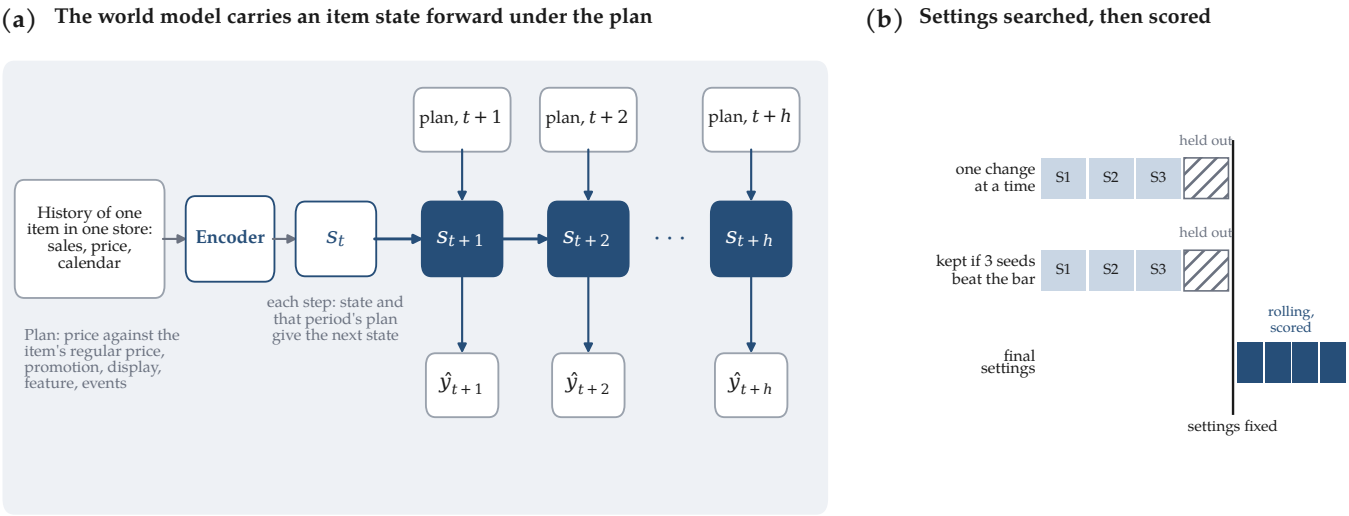


Figure 2: The world model carries each item's state forward under the plan. (**a**) An encoder turns the recent history of one item in one store into a state s_t. A transition step takes the state and one period's plan and gives the state one period later; each period's sales are read from the state as it stands after the plan so far. (**b**) Each model's settings were searched one change at a time on early windows (S1 to S3), a change kept only if its mean over three seeds beat the best so far by the keep bar, with the latest search window held out of every decision; the final settings were then scored once on the later rolling windows (Section 7).

RQ2, nature of the line. Does the forecast hold up on every kind of line? Grocery sales are concentrated: a short head of fast lines carries most of the units, and a long tail of slow lines sells a few units a week or none. Lines also differ in how much history they have. The fast lines are defined as the items that together make 80% of unit sales over the year before the forecast, the slow lines as the rest, and lines with short history as those whose first sale falls less than 26 weeks before the forecast. Each part is scored on its own series at every grain, so that the total of the fast lines, the slow lines of each store, and so on, are each judged as a planner would judge them. The share of items that makes 80% of sales is reported for each data set.

RQ3, seen and unseen data. Does the forecast hold up on items and stores the model was not trained on? Ranges change and stores open, so part of every forecast is for lines or stores with little or no history in the model. A fifth of the items, or of the stores, is kept out of training; once trained, each model reads the kept-out series' recent sales and forecasts them. The cost of not having

trained on them is the accuracy lost on that fifth against a model trained on everything.

RQ4, transferability. Do the settings carry to another retailer, and does training on other retailers' data help? Settings tuned on M5 are carried unchanged to Favorita and compared with settings tuned on Favorita itself. Each model is also trained on its data set together with the other data sets, and compared with the same model trained on its own data alone.

The broad view. The four parts are then brought together: which model, or combination, gives the best balance across grains, lines and seen and unseen data, and where each kind of model fits in the planning cycle. Findings are reported for each grain and each kind of line, since that is what a retailer needs to choose a model for a decision; the broad view simplifies them and does not replace them.

3 Related work

3.1 Retail forecasting and the promotion plan

Fildes, Ma and Kolassa [5] review retail forecasting research and practice and note that most studies forecast and evaluate at one level of aggregation, usually the item, although retail decisions are spread across levels. Sales of a single item in a single store are intermittent, which led to dedicated methods for intermittent demand [6, 7] and to evaluating forecasts as count distributions [8]. Studies of intermittent items score forecasting methods both by their error and by the stock they lead to [15, 16], and Kourentzes, Trapero and Barrow [17] propose choosing a forecasting model for its inventory cost rather than its accuracy. The Tweedie family [9], a compound Poisson–gamma distribution with a point mass at zero, is the loss many M5 entries used, and the loss we use.

Promotions move retail sales more than anything else in the plan. PromoCast forecast promoted-week sales for a chain from the planned event and the item's past response [10]. Van Heerde, Leeflang and Wittink [11] and Macé and Neslin [12] measured the dips in sales before and after a promotion, when shoppers stock up and then buy less. Ma, Fildes and Huang [13] showed that promotional information from within and across categories improves item-level forecasts. For perishable supermarket items, van Donselaar and colleagues [14] found that promotion lifts come more from substitution within the category than from stockpiling, and that forecasts improve when routine and non-routine categories are modelled apart. Gupta [18] split the promotion bump into brand switching, purchase timing and quantity, and van Heerde, Gupta and Wittink [19] showed that in units only about a third of the bump is switching from other brands. Attraction models make an item's sales the category's sales times its share [20, 21]; our brand-switching step follows that structure. Dominick's Finer Foods ran pricing experiments across its stores [22], and its scanner data are behind much of this work [23].

3.2 Decisions across grains and lines

Retail decisions read the same demand at different grains and over different spans: replenishment item by store by day, promotions item by week, category reviews by week, and range, space and supplier terms over a year or more [5]. Forecasts made separately for each grain need not agree, and the literature restores agreement by reconciliation, across the product and store hierarchy [30, 31], across levels of time [26], and across both together [27]. The alternative studied here, a single forecast at the finest grain summed upward, is coherent by construction and is the way the M5 competition scored its entries [1]; its cost is that one model must serve every grain.

Lines also differ in how much they sell and how much history they have. A small share of items carries most of the units, and the long tail sells a few units a week or none, which led to dedicated methods for intermittent demand [6, 7]. Lines with little or no history are forecast from the histories of similar products: by matching a new product to the life-cycle curves of earlier ones [28], or from product attributes [29]. Global models trained across many series [34] can forecast a series they were not trained on from its recent sales alone, which is how the models in this study meet new items and new stores.

3.3 Hierarchies, combinations and evaluation

Forecasts that must add up across a hierarchy are studied as hierarchical forecasting, from bottom-up and top-down to optimal reconciliation [25, 30, 31]. Our models forecast item by store and their forecasts are summed, as the M5 accuracy competition scored entries, with a weighted, scaled error averaged over twelve levels [1]; reconciliation is tested as a benchmark. The scaling follows Hyndman and Koehler [32], and a companion competition scored forecast distributions [2]. Tashman [33] sets out why a comparison should use several forecast origins. Hansen, Lunde and Nason's model confidence set [77] gives the set of models that cannot be told apart from the best.

Combining forecasts from different models reduces error [40, 41], and an equal-weight average is hard to beat

with estimated weights, the forecast combination puzzle, which Smith and Wallis [72] explain by the error in estimating the weights; Wang and co-authors [73] review the field. We fix an equal-weight ensemble in advance and test weights fitted on earlier windows against it. Interval forecasts by calibrating each model on its own past errors follow conformal prediction [74].

3.4 Global models, trees and neural forecasters

Global models, fitted across many series at once, are often more accurate than one model per series [34]. In M5 the leading entries were built on gradient-boosted trees [35, 36]; the winner averaged direct and recursive LightGBM models [4]. Neural global models include N-BEATS [37], DeepAR [38] and the Temporal Fusion Transformer [39], the last two taking known future inputs such as price. Graph neural networks share information between related series [42]; our item-to-item experiments use attention [67] and conditional similarity networks [43].

3.5 Pretrained foundation models

Pretrained time-series models forecast series they were not trained on. TimesFM [44] is a decoder-only transformer pretrained on a large corpus of series; Chronos [46] tokenises values and trains a language model on them, and Chronos-Bolt is its faster direct-quantile version; Lag-Llama [47], Moirai [48] and TimeGPT [49] follow similar lines. TimesFM 2.5 adds a covariate method that fits a linear model on future inputs and forecasts the remainder with the pretrained model [45]. Timestral 1.0 [76], released by Mistral on 15 October 2026, reads known future inputs inside the model itself and returns 21 quantiles. We test whether any of them, given the plan, matches models trained on it.

3.6 World models and latent prediction

A world model learns a state of its environment and how that state moves under actions, so that the effect of a planned action can be rolled forward before it is taken [50, 51, 52]. LeCun [53] proposed joint-embedding predictive architectures (JEPA), which predict the representation of the future rather than the future itself; I-JEPA [54] and V-JEPA 2 [55] apply the idea to images and video, the latter with an action-conditioned predictor used for planning. Such models guard against a collapsed representation with a slowly updated target encoder [56], a variance and covariance penalty [57], or the isotropic-Gaussian penalty of LeJEPA [58], also used in LeWorldModel [59]. TD-MPC2 [60] rolls a latent state forward under actions; DWM [61] keeps the effect of an action apart from the world's own dynamics, and AD-WM [62] trains world models to tell actions apart for counterfactual planning. Stable-worldmodel [63] evaluates world models under controlled changes, as our perturbation tests do. Closest to us, WorldTS [64] learns covariate-conditioned latent dynamics for forecasting with known future inputs. Our earlier work, M-JEPA [65], compared JEPA fine-tuning recipes with leakage-resistant splits. In retail the actions are the plan. Our world model uses a recurrent transition [66] with layer normalisation [68] on a transformer encoder [67]. It is not compared here with the same network without the transition; the carried-state tests (Section 12) show what the state does, not how much accuracy it adds over a network without one.

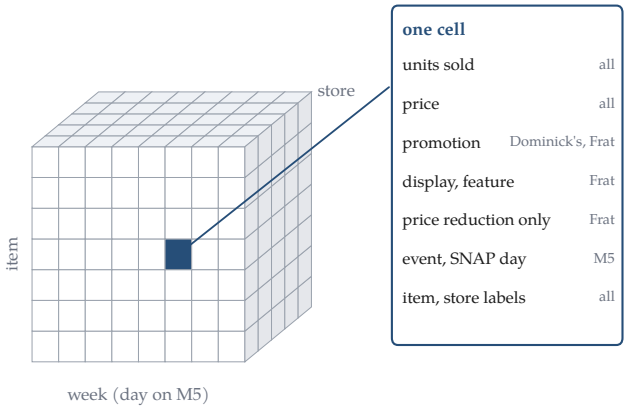
3.7 What this paper adds

Two results here are not new: an equal-weight average of different models is hard to beat [72, 73], and gradient-boosted trees are strong on retail panels [1]. The finding here is that the average of a tuned world model and tuned trees is robust across grains, but the choice of model changes for items absent from training. The paper adds four things. It scores one forecast at every grain a retailer decides at, period by period and over the whole promotion block, so that the choice of model is read grain by grain rather than from one score averaged over levels. It reads the same comparison by the kind of line, separating the fast lines that carry most sales from the slow tail and from lines with short history. It tests what each model loses on items and stores it was not trained on, and whether its settings and training data carry across retailers. And it tunes the world model and gradient-boosted trees by a common search procedure, each over its own settings, so that the comparison is between two tuned models.

4 Data

All four data sets are public (Table 1) and share one table shape (Figure 3a). *M5* holds daily Walmart sales of 3,049 items in 10 stores across three states from 2011 to 2016, with prices by week, event days and the days on which SNAP food assistance is paid [1]. *Dominick's Finer Foods* holds weekly scanner data by store for a Chicago chain from 1989 to 1997, with price, a promotion code and cost [23]. We use six categories chosen before any final run (cheese, snack crackers, soft drinks, bottled juice, analgesics, frozen entrees). *Breakfast at the Frat* holds weekly sales of 55 breakfast items in 77 stores from 2009 to 2012, with the shelf price, base price, and whether the item was on display, in the weekly circular (feature) or on a tag-only price reduction [24]. *Corporación Favorita*

(a) One row per item, store and period



(b) The levels a forecast is scored at, and who plans there

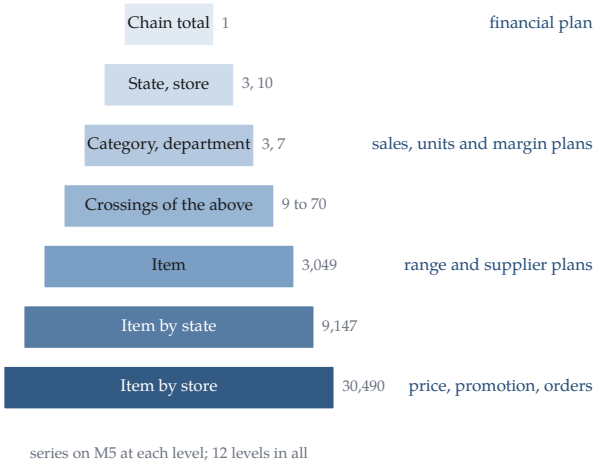


Figure 3: Every data set is one table of item, store and period, scored at every level a retailer plans at. **(a)** Each cell holds the units sold, the price and whatever plan the data set records; a lever a data set does not record is passed to the models as unknown, not as zero. **(b)** The M5 hierarchy with the number of series at each level, and the decisions usually made at each. Dominick's and Frat have six levels: total, category, store, category by store, item and item by store; Favorita has total, store, family, store by family, item and item by store.

Table 1: Four public data sets, all scored item by store and at every level above it. Rolling windows were added in this revision and precede every window used for development.

Data	Series	Plan inputs	Rolling windows
M5 (daily)	30,490	Price, event days, SNAP days	Four of 28 days, Nov 2015 to Feb 2016
Dominick's (weekly)	Six categories	Price, promotion	Six of 8 weeks per category
Frat (weekly)	3,909	Price, promotion, display, feature, price reduction only	Four of 8 weeks
Favorita (daily)	Top 10 stores	Promotion, holidays and events, paydays	Four of 28 days, the last ending 15 Aug 2017

holds daily sales of an Ecuadorian grocery chain to 15 August 2017 [75]. It was added in this revision as a data set kept out of development: the series are the 10 stores with the most units sold in 2016 crossed with the items sold in all ten that year, from 1 January 2015; returns are set to zero; the plan is the promotion flag, national holidays and events, and paydays (the 15th and the last day of each month). Favorita records no prices, so its WRMSSE weights are unit weights, not revenue weights. No raw data is redistributed.

Measure. We use the weighted root mean squared scaled error (WRMSSE) of M5 [1] at each level of the hierarchy: 12 levels on M5 (Figure 3b) and six on the others. For a level with series j, history $y_{j,1}, \dots, y_{j,T}$ before the window and forecasts $\hat{y}_{j,t}$ for the H periods after it,

$$\text{RMSSE}_j = \sqrt{\frac{\frac{1}{H}\sum_{t=T+1}^{T+H}(y_{j,t} - \hat{y}_{j,t})^2}{\frac{1}{T-t_j}\sum_{t=t_j+1}^{T}(y_{j,t} - y_{j,t-1})^2}}, \quad (1)$$

$$\text{WRMSSE} = \sum_j w_j \, \text{RMSSE}_j, \quad (2)$$

where t_j is the series' first period with sales and w_j is its share of the level's dollar sales (units on Favorita) over the last weeks of training. We report item by store, where orders and promotions are decided, the chain total, which a financial plan is set against, and the mean over all levels (the M5 score). Each Dominick's data set is one category, so its category level repeats the total and its category-by-store level repeats the store level; the results count each distinct level once, four levels on Dominick's. Lower is better.

5 Models

Table 2 lists every model. The world model and LightGBM were each tuned on every data set by a common search procedure (Section 7); their final settings are in Table 3, and the constraints each works within are in Appendix A. Each tuned model's forecast is the average of five fits with different seeds, except on M5, where each model is a single fit to bound the cost of the larger data set.

Table 2: Every forecast compared. Plan: sees the plan for the forecast periods. Trained: fitted on the data set. Tuned: settings set by the search of Section 7.

Model	Plan	Trained	Tuned	Technique
Average	yes	yes	yes	Equal-weight mean of the world model and LightGBM
Three-model ensemble	yes	yes	yes	Equal-weight mean of the world model, LightGBM and TimesFM with the plan
World model	yes	yes	yes	Item state rolled forward under the plan
LightGBM	yes	yes	yes	Direct gradient-boosted trees
Temporal Fusion Transformer	yes	yes	no	Attention over history and plan
TimesFM with the plan	yes	no	no	TimesFM 2.5 with its covariate method
Timestral with the plan	yes	no	no	Timestral-28M with known future inputs
TimesFM	no	no	no	TimesFM 2.5, 512 periods of units
Chronos-Bolt	no	no	no	Chronos-Bolt base
Seasonal naive	no	no	no	Sales one season earlier

5.1 World model

We use *world model* for an action-conditioned state model: a learned state for each item, carried forward period by period under the plan, which plays the part of the action, and read out as sales. The same model can be rolled forward under a plan that was never made, which is the question a merchant asks of it. It is trained end to end on sales. The objective that trains the state to predict its own later values, the latent term of Section 5.2 and JEPA pretraining, was tried on every data set and kept on two of the nine, Favorita and snacks (Section 7). The case for the model therefore rests on what the carried state adds to the forecast, which the carried-state tests measure (Section 12), and not on that objective.

Encoder and state. For item i in one store, an encoder (a transformer over the last L periods) reads sales and price, each scaled by the item's recent level ℓ_i, together with the calendar and the recorded levers, and gives a state s_t; the item and store labels enter the readout. The item's level is its average over the history window, or, on Dominick's, over its last 8 weeks.

Transition. A gated recurrent transition f carries the state forward under the plan p_{t+h} for each future period,

$$s_{t+h}(p) = \mathrm{LN}(f(p_{t+h},\, s_{t+h-1}(p))), \quad s_t(p) = s_t, \tag{3}$$

where LN is layer normalisation (used on M5, Dominick's and Favorita). The plan holds the period's price as a share of the item's regular price (its highest price in the last 12 periods) and the recorded levers.

Readout and loss. Expected sales are a baseline under the "nothing planned" plan p^0 (regular price, no lever) times a lift,

$$\begin{aligned}\log \mu_{i,t+h} = \log \ell_i + b(s_{t+h}(p^0), p^0_{t+h}) \\ + g(s_{t+h}(p), p_{t+h}) - g(s_{t+h}(p^0), p^0_{t+h}),\end{aligned} \tag{4}$$

with b and g small networks that also take the item's labels and the horizon step, and the log forecast bounded to $[-10, 10]$. The lift is exactly zero when nothing is planned, and the baseline does not move when the plan changes; both are checked by tests in the repository. Training minimises the loss of Section 5.2 on sales relative to ℓ_i, with batches of item, store and date drawn at random.

Brand switching (Dominick's). In a store and week where several items are promoted, much of each item's lift is taken from the others. Following attraction models [20, 21], the world model fits on the last 8 training weeks one number per run, the share α of the summed lift that the category total keeps, and rescales its item forecasts in each store and week so that the category total is the baseline total plus α times the summed lift: with baselines b_i, lifts λ_i and forecasts $\hat{y}_i = b_i + \lambda_i$ for the items i of one store and week,

$$\hat{y}'_i = \hat{y}_i \, \frac{\sum_k b_k + \alpha \sum_k \lambda_k}{\sum_k \hat{y}_k}, \tag{5}$$

so each item keeps its share of the category forecast. With Λ_g the summed lift and G_g the actual minus baseline sales of store-week group g in the last 8 training weeks, α is the least-squares slope $\sum_g \Lambda_g G_g / \sum_g \Lambda_g^2$, kept between 0 and 1. α is a forecast-implied split of lift, not an identified estimate of where promoted sales come from, because promotions are not assigned at random.

5.2 Training losses

The loss used. The world model is trained on sales relative to the item's level, $z = y/\ell_i$, by default with the Tweedie deviance for a mean μ and power $p = 1.5$,

$$L_p(z, \mu) = -\frac{z\,\mu^{1-p}}{1-p} + \frac{\mu^{2-p}}{2-p}, \tag{6}$$

the negative log-likelihood, up to terms free of μ, of a compound Poisson–gamma distribution with a point mass at zero [9]. Because the loss is on z, every item counts alike whatever its volume. The search replaced it on three data sets (Table 3): on M5 and Favorita by a hurdle negative binomial, a probability of any sale times

Table 3: The tuned settings used for every result of Section 8, read from the final configurations. Copies: forecasts averaged over that many fits with different seeds. Size: state width by encoder layers. Steps: training steps (and pretraining steps on the latent objective, where used). Periods: training periods read. The changes each search kept are listed in Table 5.

	World model					LightGBM			
Data set	Copies	Size	Steps	Periods	Loss	Copies	Rounds	Periods	Objective
M5	1	128×3	30,000	730	Hurdle negative binomial	1	800	730	Tweedie
Favorita	5	128×3	30,000	730	Hurdle negative binomial	5	800	730	Poisson
Frat	5	64×2	3,000	90	Tweedie	5	500	90	Tweedie
Analgesics	5	64×2	6,000	156	Tweedie	5	500	156	Tweedie
Bottled juice	5	64×2	6,000	156	Tweedie	5	500	156	Poisson
Cheese	5	64×2	6,000	156	Tweedie	5	500	156	Tweedie
Frozen entrees	5	64×2	6,000	156	Tweedie	5	500	156	Tweedie
Soft drinks	5	64×2	6,000	156	Tweedie	5	500	156	Tweedie
Snacks	5	64×2	6,000, pretrained 18,000 steps	156	Negative binomial	5	1000	156	Tweedie

a negative binomial for the units sold when there is one, and on snacks by a negative binomial; in each the spread of the negative binomial is learnt for each item. LightGBM uses the Tweedie deviance with $p = 1.1$ on units, so large items count for more, or the Poisson deviance where its search chose it, and the Temporal Fusion Transformer a negative binomial likelihood on units.

The changes tried. Each change below adds to or replaces L_p for one batch B of item, store and period examples. Table 4 scores each against the configuration it changed in the first version, where none was kept; the search of Section 7 tried them again on the tuned settings, with the likelihoods above, and kept the changes listed in Table 5.

- *Negative binomial* on units, $-\log \mathrm{NB}(y;\, \mu\ell_i, r)$ with the spread r learnt.
- *Totals term*: the loss adds $\lambda_T \,\mathrm{mean}_t[(\sum_{b\in B_t} \hat{y}_b - \sum_{b\in B_t} y_b)/\max(1, \sum_{b\in B_t} y_b)]^2$, the relative gap between summed forecasts and sales over the examples B_t sharing a date, so that a small bias in the same direction on many items is penalised ($\lambda_T = 1$).
- *Volume and dollar weights*: each example's loss times ℓ_i^γ, or $(\ell_i \bar{\pi}_i)^\gamma$ with $\bar{\pi}_i$ the item's usual price, normalised to mean one in the batch, with $\gamma = 0.9$ and $p = 1.1$, which makes the loss close to LightGBM's on units or dollars.
- *Peak weight*: each period's loss times $1 + \kappa \min(|z_t - 1|, 5)$, normalised to mean one, so that peaks and dips count for more ($\kappa = 0.5$, with the dollar weight above).
- *After-promotion weight*: each period with no lever on that follows a period with one counts $1 + w$ times, normalised to mean one ($w = 2$ or 5).
- *Latent term*: from the state now and the plan, the transition predicts the state the encoder gives once those periods are history, against a slowly updated copy of the encoder; the loss adds $\lambda_Z \sum_h \delta^h \|s_{t+h}(p) - \bar{s}_{t+h}\|^2$ with $\delta \le 1$ discounting far periods, and optionally a variance and covariance penalty on the state [57]. It was also used alone to pretrain the encoder, which was then frozen, trained at a lower rate or trained fully on the forecast loss.

5.3 Average and ensemble

The average is the equal-weight mean of the tuned world model's and the tuned LightGBM's forecasts for each item, store and period, with no weights fitted. The three-model ensemble adds TimesFM with the plan, also with equal weights; it was fixed on 13 October 2026, before any result of the revision, and is reported beside the average. The forecast by history of Section 8.6 uses the world model alone for series with less than 26 weeks of sales and the average for the rest.

5.4 Comparison models

The Temporal Fusion Transformer and the pretrained models are untuned reference models: the Temporal Fusion Transformer keeps the settings of the first version, and the pretrained models run as published. Searching each of them on nine data sets as the two tuned models were searched was beyond the compute of this study, so the comparisons that decide the rule of Section 8.6 are between the two tuned models and their average. *LightGBM* [35]: a direct model, one per store on M5 and Favorita, reading sales at lags no shorter than the horizon, rolling averages and spread of sales, price against the

Table 4: No change to the training loss became a default: each was lower in error than the configuration it changed in at most half its comparisons. Each row is one change to the Tweedie loss of the world model or lift model, scored on the development windows against the configuration with the same data, windows and every other setting. WRMSSE: overall (mean of the hierarchy levels), averaged over the seeds and windows both ran and over the data sets in the row. Lower: the number of data sets (or category and model pairs) in which the change had the lower error. The weight after a promotion comes from a separate rerun of the world model in the same job. Rows marked not run were configured but have no full-data run in the records. Built by `scripts/loss_ledger.py`.

Change to the loss	Tested on	With it	Without	Result
Negative binomial likelihood	Dominick's, Frat	0.537	0.541	Lower in 1 of 2
Totals term	Dominick's, Frat	0.655	0.583	Lower in 0 of 3
Unit-volume weight and Tweedie power 1.1	Dominick's	n/a	n/a	Not run on full data
Dollar-volume weight and Tweedie power 1.1	Dominick's	n/a	n/a	Not run on full data
Peak weight, dollar weight and Tweedie power 1.1	Dominick's	0.853	0.856	Lower in 5 of 12
Weight 3 on the periods after a promotion	Dominick's	0.834	0.832	Lower in 2 of 6
Weight 6 on the periods after a promotion	Dominick's	0.810	0.832	Lower in 2 of 6
Latent term, weight 0.1	Frat	0.411	0.406	Lower in 0 of 1
Latent term, weight 0.5	Frat	0.414	0.407	Lower in 0 of 2
Latent term, weight 2	Frat	0.410	0.406	Lower in 0 of 1
Latent term with a spread penalty and discount	Dominick's, Frat	0.567	0.549	Lower in 0 of 4
Latent pretraining, encoder frozen afterwards	Frat	0.428	0.406	Lower in 0 of 1
Latent pretraining, encoder full afterwards	Frat	0.407	0.406	Lower in 0 of 1
Latent pretraining, encoder at a lower rate afterwards	Frat	0.411	0.407	Lower in 0 of 3
Latent pretraining, then forecast with the latent term	Frat	0.415	0.406	Lower in 0 of 1
Latent pretraining, then forecast with a totals term	Dominick's, Frat	0.729	0.667	Lower in 0 of 1

item's highest recent price, the calendar and the plan inputs, with the item as a label; tuned as in Table 3. *Temporal Fusion Transformer* [39]: one model per data set on Dominick's and Frat, fitted on the last 156 weeks with the plan as known future inputs and a negative binomial likelihood, 1,500 steps; it was not run on M5, where its training cost was beyond the compute available, nor on Favorita, whose model list was fixed without it. Chronos-Bolt and TimesFM from history alone were not run on Favorita either. *TimesFM* [44, 45]: the 200M-parameter TimesFM 2.5 checkpoint given each series' last 512 periods; we use its mean output, chosen in the first version because its median forecast fell well below M5 totals, as the median of intermittent daily sales does. *TimesFM with the plan*: the same model through its covariate method, a linear model on the plan inputs fitted over groups of 2,048 series, then TimesFM on what that model leaves. *Chronos-Bolt* [46]: the base checkpoint, zero-shot, the mean of its quantiles. *Timestral* [76]: Timestral-28M (code commit a534bdb), zero-shot, with the plan as known future inputs (price relative to the context's highest price, other levers as given) and from history alone; the point forecast is the mean of its 21 quantiles, each weighted by the share of probability nearest it. *Seasonal naive*: sales one season earlier.

6 Experimental setup

Windows. Every result of Section 8 is scored on rolling windows that no choice of setting ever saw: six 8-week windows on each Dominick's category, four on Frat, and four 28-day windows on M5 (November 2015 to February 2016) and on Favorita (the last ending on its last day), each forecast from everything earlier. The searches of Section 7 scored windows before these, and held the latest of their own windows out of every decision. The windows used to develop the first version of the models lie after the rolling windows and are not scored in this paper (Figure 1a).

Rules fixed in advance. The revision plan, with every window, model, test and pass rule, was written down before the revision's first run, and every change to it is a dated entry, made before the results it governs were read unless the entry says otherwise. Each data set's world model and LightGBM run in every rolling window with the final settings of their searches (Table 3), unchanged from window to window; each forecast is the average of five fits with different seeds (a single fit on M5). No result on the rolling windows changed a setting, and every model, window and test is reported.

Comparing two models. For models A and B, window k and a level, let $d_k = \mathrm{WRMSSE}_k(A) - \mathrm{WRMSSE}_k(B)$. Within a window, d_k gets a 95% interval from 1,000 resamples of items: an item drawn twice counts twice in every series and aggregate it feeds, with the weights recomputed. Across the K windows, the mean $\bar{d} = \frac{1}{K}\sum_k d_k$ gets a 95% interval from the 2.5% and 97.5% quantiles of $\bar{d}^*$ over 5,000 resamples of windows with replacement. A model is said to beat another only when that interval excludes zero; when it does not, we say no difference was detected, which is not a finding that the two are equal. For each data set and level we report the 90% model confidence set [77]: starting from all models $\mathcal{M}$, the range statistic $T = \max_{a,b\in\mathcal{M}} |\bar{d}_{ab}|/\hat{\sigma}_{ab}$, with $\hat{\sigma}_{ab}$ from the bootstrap over windows, is compared with its bootstrap distribution under equal expected loss (5,000 resamples); while the p-value is below 0.10 the model whose standardised losses against the others are largest is removed, and the models left form the set. With four to six windows these tests have little power, and adjacent windows are dependent: each is forecast from all data before it, so they share most of their training history, and an error tied to a season or a stretch of trading can recur in neighbouring windows. A bootstrap that resamples whole windows treats them as exchangeable and so understates that dependence; we therefore also count, over levels and windows, the comparisons in which one model is clearly more accurate than the other by the item bootstrap (Section 8.1).

Intervals. Each model's interval forecasts come from its own errors in the earlier rolling windows of the same data set, a rolling form of conformal calibration [74]. With $e_{j,h} = (y_{j,t+h} - \hat{y}_{j,t+h})/\sigma_j$ the scaled errors of the earlier windows at horizon h (σ_j the series' one-step scale), the interval at level $1-\alpha$ is $[\hat{y}_{j,t+h} + \sigma_j q_{\alpha/2,h},\ \hat{y}_{j,t+h} + \sigma_j q_{1-\alpha/2,h}]$, with $q_{\cdot,h}$ the empirical quantiles of the pooled $e_{\cdot,h}$. Where a level has fewer than 50 earlier errors for each horizon, all horizons are pooled into one set of quantiles, and interval ends are rounded outwards to whole units except on Favorita; this fix was added on 16 October after the first intervals covered far less than planned at the totals, and is used throughout. The first rolling window has no earlier window and is not scored. We report the scaled pinball loss averaged over the quantiles, weighted as in the M5 uncertainty competition [2], and coverage at 50%, 80% and 95%. Models that give quantiles of their own (Chronos-Bolt, TFT) are also scored on them, over every window.

Decisions. Three simulations turn forecasts into costs. *Store orders*: for each item-store-period, order the model's interval quantile at a critical ratio r of 0.5, 0.8 or 0.95; each unit over costs 1 and each unit short $r/(1-r)$. *Category plan*: for category-by-store nodes n with sales $y_{n,t}$, forecast $\hat{y}_{n,t}$ and average price $\bar{\pi}_n$ before the window,

$$C = \frac{\sum_n \bar{\pi}_n \sum_t |y_{n,t} - \hat{y}_{n,t}|}{\sum_n \bar{\pi}_n \sum_t y_{n,t}}, \tag{7}$$

the absolute error in dollars over sales in dollars. *Promotion decisions*: in the generated store worlds of the companion paper [71], where each promotion's true lift is known, promote when the predicted lift times margin exceeds a fixed cost; regret is the value lost against deciding with the true lift. The first two are reported against seasonal naive. All are simulations under stated costs, not business results.

Reconciliation. Base forecasts at every node come from AutoETS fitted on the history before each window (seasonal at the aggregate nodes, non-seasonal item by store). They are reconciled bottom-up, top-down by historical proportions, and by the MinT family [31]. With S the summing matrix (nodes by item-store series) and $\hat{\mathbf{y}}$ the base forecasts at every node, the reconciled item-store forecasts are

$$\tilde{\mathbf{b}} = (S^\top W^{-1} S)^{-1} S^\top W^{-1} \hat{\mathbf{y}}, \tag{8}$$

with $W = I$ (ordinary least squares), $W = \mathrm{diag}(S\mathbf{1})$ (structural weights), W the diagonal of in-sample error variances, or $W = \lambda\mathrm{diag}(\hat{\Sigma}) + (1-\lambda)\hat{\Sigma}$ with the Schäfer–Strimmer intensity λ (MinT-shrink, where the hierarchy has at most 6,000 nodes: Frat); nodes with no in-sample error are given the smallest variance of the others. Each model's own item-by-store forecasts are also combined with ETS above them by weighted least squares.

Ranking. Accuracy says how close a forecast is, not whether it puts items in the right order. For every window we report the Spearman correlation, across item-store series that sold in the window before, between the forecast change (horizon total over the previous horizon's total) and the actual change; the share of the tenth of nodes forecast to grow most that are in the actual top tenth (0.10 by chance); the Spearman correlation across promoted periods between forecast and actual units over the series' unpromoted level; and bias, forecast over actual. These were written down before they ran and have no pass rule.

Analyses added after the first results. Four analyses were added after the first scores of the tuned models were read, each written down before it ran: the scores by line type (Section 8.2); the whole window scored as one block; the average and the forecast by history on the

transfer test's kept-out series (Section 8.3); and the supporting analyses of Sections 9 to 14 rerun on the tuned models. The transfer test and the pooled training of RQ4 were written down before any tuned model was scored.

7 Searching the settings of both models

Both models were tuned with the same procedure on every data set. On the windows the searches scored, tuning improved LightGBM on all nine data sets and the world model on six, and the gains were mostly small next to the differences between data sets. The held-out window agreed with the search windows only weakly, so the paper judges the models on later windows the searches never saw. Table 5 gives every search from its starting settings to its final settings, and Figure 4 shows every run in the order it was made.

7.1 The procedure

The procedure is a fixed loop, written down before it ran, that changes one setting at a time. It starts from the settings the first version of this paper used. Each change on a menu fixed in advance is run once, on one seed, on top of the best settings so far. Every change that beats the best settings on that seed is run on two more seeds. The best of those is kept if its mean over three seeds beats the best settings' mean by at least the keep bar, which is the spread of the three seeds of the starting settings. The loop then starts a new round from the new best and stops when a round keeps nothing. A change cannot be kept because one seed was lucky, and the bar is set by the model's own noise on that data set before any change is tried.

The windows the loop scores lie before every window this paper reports. The latest of them is held out: it is scored for every three-seed run and never used to decide. The menus differ between the models, because the models have different settings, but each covers what a practitioner would try first: for the world model the loss weights, the roll-forward, training length, learning rate and batch, weight averaging and the latent checks of Section 5; for LightGBM the number of leaves, the learning rate and number of rounds, the minimum data in a leaf, the objective, feature sampling and the lag windows.

After its loop each world model had two more rounds of one change each. The first tried four count likelihoods that treat weeks with no sales on their own: the negative binomial, the same with a spread for each item, the zero-inflated negative binomial and the hurdle negative binomial. The second tried JEPA pretraining of the encoder with the forecast head fitted afterwards at a lower rate, at two lengths of pretraining. Both used the keep rule above. LightGBM has no counterpart to either, so it had none. Pretraining used only each data set's own history, so the two models still saw the same data.

Each model on each data set was searched as a five-copy average, the mean forecast of five fits with different seeds, except on M5, where both models were single fits. Section 7.4 shows why the average matters.

7.2 What the searches kept

The searches made 1,029 runs, 472 of the world model and 557 of LightGBM, and kept 23 changes, 10 for the world model and 13 for LightGBM. LightGBM kept at least one change on every data set and the world model kept nothing on three of the six Dominick's categories. No single change helped the world model everywhere (Figure 5). Weighting the loss toward high-volume series made the first seed worse on most data sets. The seven-step latent check, kept on Favorita, was among the changes that moved Favorita most and did little elsewhere.

On the search windows the world model improved on six of the nine data sets and LightGBM on all nine (Figure 8). The largest gain of either model was LightGBM on Favorita, from 0.922 to 0.580. The world model's largest was on Favorita, from 0.547 to 0.483.

7.3 Did the held-out window agree?

Only partly. Across every three-seed run, the change on the search windows and the change on the held-out window are correlated at 0.29 for the world model and 0.05 for LightGBM (Figure 6). A change that helps on the search windows is a weak guide to the next window. The world model's final settings were worse than its starting settings on the held-out window on two data sets, most on snacks (0.568 to 0.608), and LightGBM's on five, most on soft drinks (0.435 to 0.507). This is why the paper reports the final settings on windows that the searches never saw, in Section 8, and not the search scores.

7.4 Seed noise, one copy against five

A single world-model fit is noisy enough to hide most of the changes the loop tries. On the six Dominick's categories the three seeds of the starting settings differ from their mean by a standard deviation of 6.3% for a single fit and 1.8% for a five-copy average (Figure 7). The single-fit search on cheese had a bar of 0.040 and kept nothing; with the average the bars on the Dominick's categories were between 0.004 and 0.034. LightGBM's seeds differ much less, and its bars were between 0.002 and 0.010.

Table 5: Every search, from starting to final settings. Scores are WRMSSE (lower is better), the mean of three seeds on the search windows and one score on the held-out window, which never decided anything. The keep bar is the seed spread of the starting settings; a change was kept only when its three-seed mean beat the best so far by at least the bar. Tries counts every run, first-seed screens and three-seed confirmations.

Data	Model	Tries	Keep bar	Starting		Final		Kept, in order
				Search	Held-out	Search	Held-out	
M5	World model	59	0.013	0.570	0.550	0.533	0.534	feedback, hurdle nb item spread
	LightGBM	51	0.003	0.603	0.553	0.556	0.522	num leaves quarter
Favorita	World model	60	0.028	0.547	0.600	0.483	0.560	latent overshooting, hurdle nb item spread
	LightGBM	99	0.009	0.922	0.576	0.580	0.573	min data in leaf 4x, objective poisson, learning rate half
Frat	World model	53	0.028	0.588	0.591	0.547	0.608	batch half
	LightGBM	52	0.003	0.501	0.546	0.489	0.546	learning rate double
Analgesics	World model	35	0.004	0.509	0.619	0.509	0.619	nothing
	LightGBM	52	0.010	0.548	0.757	0.528	0.764	num leaves quarter
Bottled juice	World model	30	0.028	0.587	0.637	0.587	0.637	nothing
	LightGBM	67	0.008	0.708	0.582	0.674	0.593	objective poisson, feature fraction 0.8
Cheese	World model	33	0.034	0.392	0.714	0.392	0.714	nothing
	LightGBM	49	0.002	0.352	0.833	0.346	0.839	longer windows
Frozen entrees	World model	95	0.007	0.246	0.251	0.216	0.241	total weight 0.1, batch half
	LightGBM	53	0.010	0.222	0.183	0.181	0.195	num leaves 4x
Soft drinks	World model	65	0.020	0.594	0.579	0.572	0.522	lr double
	LightGBM	80	0.004	0.554	0.435	0.532	0.507	num leaves quarter, max bin 255
Snacks	World model	42	0.018	0.439	0.568	0.400	0.608	negative binomial item spread, jepa hybrid long
	LightGBM	54	0.006	0.358	0.425	0.351	0.424	rounds double

8 Results

The results follow the four research questions. Every number comes from the tuned world model and the tuned LightGBM, each set by a common search procedure on each data set (Section 7), scored on the paper's windows. The average is the equal-weight mean of the two forecasts.

8.1 RQ1: grain

Use the average at every grain. Across the nine data sets and the three views of Table 6 (store grain and above by period, item grain by period, and the whole window as one block), the average has the lowest error in 22 of 27 cases and is never the least accurate of the three. The world model is the least accurate in 16 cases and LightGBM in 11.

The paired tests give the same answer. They compare two forecasts level by level and window by window, and count a difference only when it holds after resampling items. Against the world model, the average is clearly more accurate in 109 of 240 comparisons and clearly less accurate in 22; against LightGBM, 106 and 18. At item grain it is clearly less accurate than either model in at most 8 of 100 comparisons. Across data sets and levels it is in the 90% model confidence set in 39 of 48 cases, against 27 for LightGBM and 25 for the world model. The paired item tests are the main evidence: the window bootstrap behind the confidence set treats adjacent windows, which share most of their training data, as independent, and so overstates its confidence (Section 6).

Grain does not decide between the two models on their own. The world model is clearly more accurate than LightGBM in 33 comparisons at store grain and above and clearly less accurate in 33; at item grain the counts are 24 and 30. Neither is ahead at either grain across the data sets, so a retailer that picks one model for a grain is picking by data set, not by grain. The average removes that choice.

On the whole promotion block, the average is again the safe choice. It is lowest on five of nine data sets and second on the other four. LightGBM is lowest on analgesics, bottled juice and snacks, but drifts in total where the world model does not: its block error on M5 is 0.953, against 0.598 for the world model and 0.589 for the average, and the same pattern shows on Frat, Favorita and

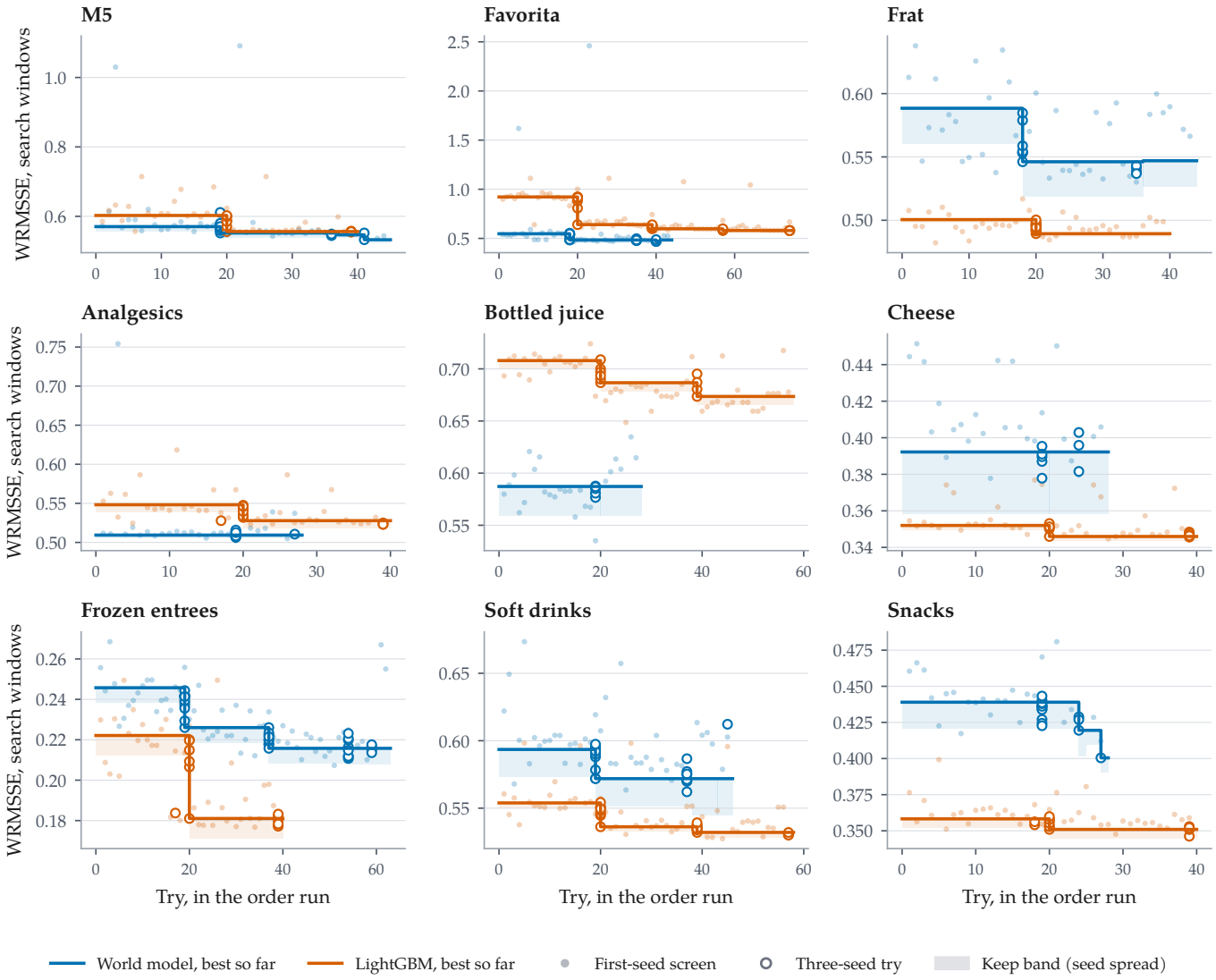


Figure 4: Every run of every search, in the order it was made. Faint dots are first-seed screens, open circles are three-seed confirmations, and the step line is the best settings so far. The band under the line is the keep bar: a change had to land below it to be kept. The world model's line runs on into its likelihood and JEPA rounds after its own loop ends.

soft drinks.

8.2 RQ2: nature of the line

Sales are concentrated, and the average holds up on both the fast lines and the slow tail; where history is short, the world model is ahead of LightGBM. On the six Dominick's categories, 11 to 21% of the items make 80% of unit sales over the year before the forecast; on M5, Frat and Favorita the head is flatter, at 37 to 42% (Table 7). On the fast lines the average has the lowest error on eight of nine data sets, and on the slow lines on eight; on snacks LightGBM is lowest on the fast lines, and on cheese the world model is lowest on the slow lines.

On series whose first sale falls less than 26 weeks before the forecast, the world model's error is lower than LightGBM's on seven of nine data sets, frozen entrees and Frat being the exceptions: for example 0.539 against 0.667 on Favorita and 0.866 against 1.024 on M5. These series are 11 to 21% of the series on the Dominick's categories, where items enter and leave the range over the years, and under 2% on M5, Frat and Favorita.

8.3 RQ3: seen and unseen data

On items the models were not trained on, use the world model alone rather than the average with LightGBM fitted with the item label; on stores they were not trained on, the world model and the average are level, and the average serves. With a fifth of the items kept out of training, the world model's error on them rises by 0.008 to 0.040, against 0.058 to 0.311 for LightGBM (Table 8). The tuned LightGBM reads the item label as a feature, so on a kept-out item it meets a label it never

Table 6: RQ1, grain. WRMSSE of the tuned world model (WM), the tuned LightGBM (LGB) and their equal-weight average (Avg), averaged over the levels at store grain and above, over the item levels, and over every level for the whole forecast window summed as one block; mean over the paper's windows. Shading marks the lowest of the three: world model, LightGBM, average.

	Store and above			Item levels			Whole block		
Data set	WM	LGB	Avg	WM	LGB	Avg	WM	LGB	Avg
Analgesics	0.688	0.698	**0.678**	0.710	0.705	**0.695**	0.552	**0.499**	0.508
Bottled juice	0.569	0.566	**0.539**	0.581	0.596	**0.563**	0.540	**0.434**	0.457
Cheese	0.381	0.471	**0.377**	0.623	0.594	**0.578**	0.482	0.444	**0.416**
Frozen entrees	0.339	0.327	**0.320**	0.375	0.350	**0.344**	0.362	0.337	**0.327**
Soft drinks	0.822	0.805	**0.754**	0.589	0.555	**0.547**	0.625	0.641	**0.523**
Snacks	1.241	**1.040**	1.071	0.545	0.516	**0.508**	0.776	**0.571**	0.612
Frat	0.471	0.479	**0.449**	0.528	0.524	**0.508**	**0.345**	0.406	0.364
M5	0.660	0.634	**0.591**	0.859	0.886	**0.856**	0.598	0.953	**0.589**
Favorita	0.314	0.325	**0.278**	0.655	0.696	**0.653**	0.482	0.565	**0.412**

Table 7: RQ2, nature of the line. Share of items that make 80% of unit sales over the year before the forecast, and WRMSSE averaged over every level, by period, on the fast lines (those items), the slow lines (the other items) and the series with short history (first sale less than 26 weeks or 182 days before the forecast). WM, LGB and Avg as in Table 6. Shading marks the lowest of the three: world model, LightGBM, average.

	Items making	Fast lines			Slow lines			Short history		
Data set	80% of units	WM	LGB	Avg	WM	LGB	Avg	WM	LGB	Avg
Analgesics	17%	0.754	0.748	**0.738**	0.756	0.769	**0.748**	**1.291**	1.459	1.350
Bottled juice	21%	0.594	0.606	**0.571**	0.513	0.516	**0.494**	**1.303**	1.393	1.312
Cheese	17%	0.505	0.507	**0.461**	**0.515**	0.584	0.526	**1.222**	1.411	1.284
Frozen entrees	19%	0.327	0.310	**0.306**	0.461	0.412	**0.402**	1.473	1.471	**1.336**
Soft drinks	11%	0.640	0.602	**0.583**	0.604	0.651	**0.595**	1.379	1.397	**1.334**
Snacks	17%	0.650	**0.496**	0.524	1.189	1.154	**1.144**	**4.204**	4.565	4.328
Frat	42%	0.432	0.442	**0.414**	0.659	0.626	**0.622**	2.703	2.642	**2.608**
M5	37%	0.717	0.700	**0.662**	0.735	0.748	**0.702**	**0.866**	1.024	0.907
Favorita	42%	0.417	0.442	**0.395**	0.496	0.522	**0.472**	0.539	0.667	**0.525**

saw; it under-forecasts these items, at 66 to 93% of actual sales, and that pulls the average down to 78 to 96%; the world model forecasts 89 to 104%. On the kept-out items the world model alone is clearly more accurate than the average on five of five data sets tested.

With a fifth of the stores kept out, the world model is clearly more accurate than LightGBM on four of five data sets and level on M5. Against the average it has the lower error on three of five but is clearly more accurate only on Frat; the average is clearly more accurate on cheese, and the two are level on snacks, M5 and Favorita.

8.4 RQ4: transferability

In the one transfer between retailers tested, M5's settings carried to Favorita, and training on other retailers' data did not help either model. M5's tuned settings carried unchanged to Favorita give 0.447 overall for the world model, against 0.428 with Favorita's own settings, and 0.450 for LightGBM, against 0.448 (Table 9); at item by store the difference is 0.002 for the world model and 0.001 for LightGBM. Training each model on its data set together with other data sets lowers the error by no more than 0.003 overall on any data set: by 0.002 for the world model on analgesics and by 0.003 for LightGBM on frozen entrees. Elsewhere it leaves the error unchanged or raises it, by as much as 0.143 for the world model on soft drinks and 0.215 for LightGBM on analgesics.

8.5 Ablation, occlusion and calibration

Each tuned model earns its place in the combination, both read the plan, and the forecast ranges hold what they state item by store. Dropping the world model from the three-model ensemble makes the item-by-store forecast less accurate on nine of nine data sets, and drop-

Table 8: RQ3, seen and unseen data. A fifth of the items, or of the stores, is kept out of training. WRMSSE on that fifth for the world model (WM), LightGBM (LGB) and their average (Avg); the cost of not training on them, which is that WRMSSE minus the WRMSSE on the same fifth of the same forecast trained on everything; and forecast over actual on the fifth. Means over windows. The last two columns name the forecast that is clearly lower on the kept-out fifth when a 90% bootstrap interval over windows of the difference excludes zero. Shading marks the lowest of the three: world model, LightGBM, average.

		WRMSSE, kept out			Cost of not training on them			Forecast over actual			Clearly lower	
Data set	Kept out	WM	LGB	Avg	WM	LGB	Avg	WM	LGB	Avg	WM or LGB	WM or Avg
Cheese	Items	**0.678**	0.731	0.687	0.011	0.058	0.035	0.99	0.93	0.96	World model	World model
Cheese	Stores	0.635	0.657	**0.612**	0.001	0.047	0.014	1.05	0.94	1.00	World model	Average
Snacks	Items	**0.626**	0.753	0.663	0.012	0.162	0.075	0.89	0.68	0.78	World model	World model
Snacks	Stores	**0.611**	0.700	0.621	0.001	0.102	0.034	1.00	0.72	0.86	World model	Neither
Frat	Items	**0.546**	0.843	0.654	0.008	0.311	0.135	0.95	0.66	0.81	World model	World model
Frat	Stores	**0.590**	0.779	0.634	0.007	0.205	0.071	0.99	0.77	0.88	World model	World model
M5	Items	**0.862**	0.918	0.871	0.040	0.083	0.053	1.04	0.84	0.94	World model	World model
M5	Stores	0.874	0.886	**0.869**	0.022	0.020	0.020	0.99	0.93	0.96	Neither	Neither
Favorita	Items	**0.709**	0.899	0.768	0.010	0.171	0.070	1.01	0.77	0.89	World model	World model
Favorita	Stores	**0.729**	0.766	0.731	0.020	0.017	0.020	0.97	0.98	0.97	World model	Neither

Table 9: RQ4, transferability. WRMSSE overall / item by store of each tuned model trained on its own data set and trained on it together with other data sets (each Dominick's category with the other five and Frat; Frat with the six Dominick's categories; M5 with Favorita and Favorita with M5). The last row carries M5's tuned settings unchanged to Favorita, to be read against Favorita's own-data row.

	World model		LightGBM	
Data set	Own data	Pooled	Own data	Pooled
Analgesics	0.699 / 0.669	0.697 / 0.671	0.702 / 0.657	0.916 / 0.723
Bottled juice	0.575 / 0.615	0.607 / 0.610	0.581 / 0.626	0.597 / 0.653
Cheese	0.502 / 0.639	0.500 / 0.632	0.532 / 0.618	0.592 / 0.650
Frozen entrees	0.357 / 0.422	0.363 / 0.431	0.338 / 0.396	0.335 / 0.406
Soft drinks	0.705 / 0.600	0.849 / 0.747	0.680 / 0.570	0.750 / 0.594
Snacks	0.893 / 0.579	0.912 / 0.572	0.778 / 0.565	0.835 / 0.610
Frat	0.490 / 0.568	0.492 / 0.570	0.494 / 0.563	0.548 / 0.626
M5	0.709 / 0.838	0.733 / 0.840	0.697 / 0.858	0.851 / 0.876
Favorita	0.428 / 0.690	0.448 / 0.699	0.448 / 0.722	0.459 / 0.738
Favorita, M5's settings	0.447 / 0.692		0.450 / 0.723	

ping LightGBM on eight (Section 10). Switching the plan off makes the world model less accurate on nine of nine data sets and LightGBM on nine; on the Dominick's categories and Frat, price and the promotion flag carry it, and events, SNAP days and paydays change the error by at most 0.019 (Section 11). The forecast ranges, built from each model's errors in earlier windows, hold about what they state item by store and are too narrow at the store; on the continuous ranked probability score the average is the most accurate item by store on nine of nine data sets. None of the models forecasts the chance of selling nothing better than each series' own past share of zero-sale periods, except on Frat (Section 13).

8.6 The broad view

Brought together, the four answers give one rule. **Forecast every item in every store as the equal-weight average of a tuned world model and tuned gradient-boosted trees, and sum it to every grain; for an item the models have not been trained on, use the world model alone until they are retrained with it.** On the data sets studied, the average is the most accurate choice at store grain and above on eight of nine data sets, at the item levels on all nine, and on the fast lines and the slow tail on eight of nine. The two models on their own are close on items they were trained on; they part company on items they were not, where LightGBM under-forecasts and the world model does not. The trigger for the excep-

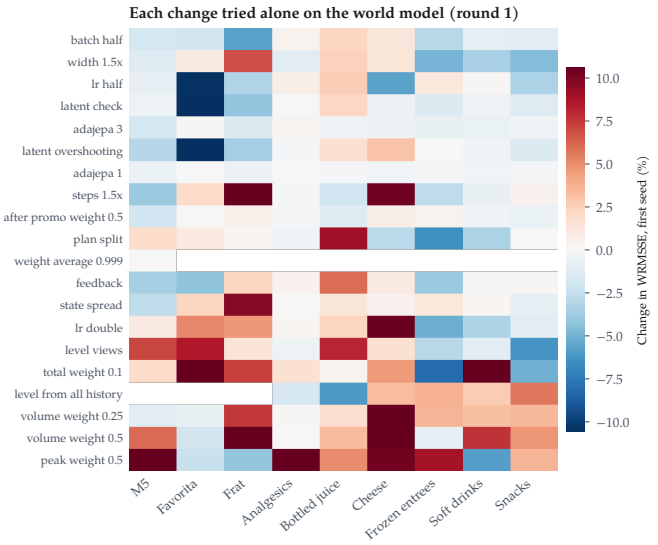


Figure 5: Each change tried alone on the world model in the first round, against that data set's starting settings on the first seed. Blue is better. A blank cell is a change that was not on that data set's menu.

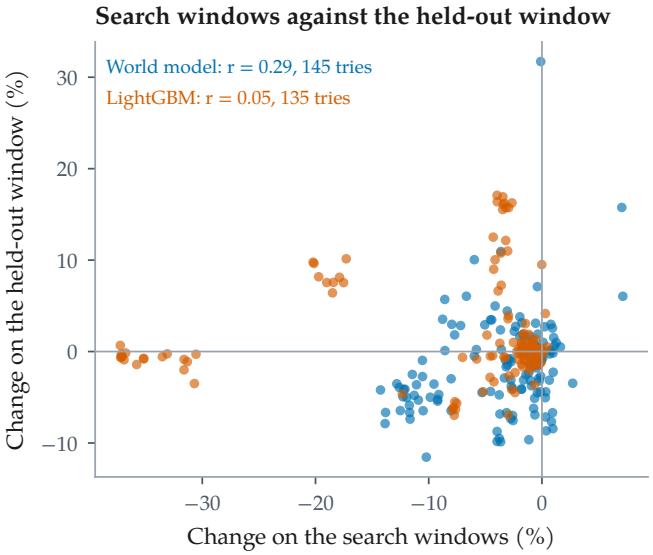


Figure 6: Each three-seed run's change on the search windows against its change on the held-out window, both against the starting settings of its search.

tion is one a merchant always knows: whether the item is new to the range. In the one transfer between retailers tested, M5's settings carried to Favorita, and training on other retailers' data did not help. Where the forecast is used only for store orders, adding TimesFM given the plan as a third member makes item-by-store forecasts a little more accurate (Section 10), at the cost of a third model to run.

Two parts of this rule were reached differently, and the difference matters to a reader who would use it. Averaging models was the recommendation of the first version, and the average of the two tuned models holds it

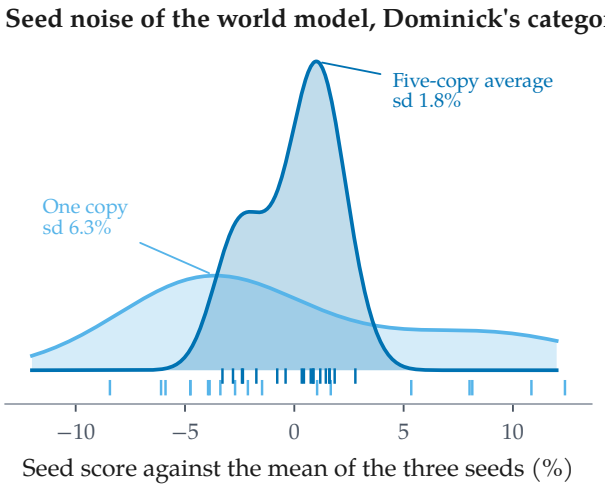


Figure 7: The starting settings' three seed scores on each Dominick's category, as a share of their mean, for a single world-model fit and for the five-copy average. Ticks are the scores.

up here. The exception was tested as part of a wider rule written down before it ran, which also gave the world model every series with less than 26 weeks of history; that half added nothing on the data with history (it was level with the average on eight data sets and worse on snacks, 0.855 against 0.790), and only the new-item half held. The rule as stated is therefore drawn from these results.

The rule is a finding on these nine data sets, not a ranking of methods for any retailer. A retailer, or a researcher, adopting the framework should rerun the full set of comparisons on its own data, including the models and combinations that lost here: the pretrained models, the Temporal Fusion Transformer, the three-model ensemble, pooling across retailers and the forecast by history. These results do not settle their order on data with other levers, other lines and other history; the code to do so is released with the paper. Section 17 sets out what the study does not cover, including forecasts longer than 8 weeks or 28 days.

9 Every model compared

Every analysis in this and the next five sections (the ablation, the occlusion, the carried-state tests, the calibration and reliability of the forecast ranges, and the decision costs) was rerun on the tuned models of Table 3, on the same rolling windows as Section 8; none of its numbers comes from the settings of the first version.

The tuned pair and their average are more accurate than every model not tuned on the data, on every data set but one. Table 10 adds the pretrained models, the Temporal Fusion Transformer and seasonal naive. TimesFM given the plan is the most accurate of the mod-



Figure 8: Each model's starting and final settings on each data set. Open circles are the starting settings and filled circles the final settings, on the search windows (top) and the held-out window (bottom).

els not trained on the data on eight data sets. On Frat it is more accurate than either tuned model alone (0.482 against 0.490 for the world model and 0.494 for LightGBM) but not than their average (0.469). Timestral given the plan forecasts far above sales on the weekly data, where a few extreme upper quantiles dominate the mean of its output: 13.892 on analgesics against 0.723 for TimesFM given the plan.

10 Ablation: what each model adds to the combination

Each tuned model makes the combination more accurate, and TimesFM adds a little at item by store. Table 11 drops each member of the three-model ensemble in turn. Without the world model the ensemble is less accurate item by store on nine of nine data sets, and clearly less accurate in 22 of 48 window comparisons (clearly more accurate in 2); without LightGBM, in 34 (clearly more accurate in 6). Without TimesFM, which leaves the average of the two tuned models, it is clearly less accurate item by store in 16 comparisons and clearly more accurate in 5. Item by store, then, adding TimesFM with the plan to the average helps; over all levels the average is in the model confidence set more often (Section 8.1). A retailer whose main use is store orders can add it at the cost of a third model to run. Weights fitted on earlier windows to the item-by-store error beat equal weights in 14 of 39 comparisons and lost in 7; weights fitted to the error over all levels beat them in 8 and lost in 16.

Figure 9a shows the same comparison data set by data set. Dropping the world model raises the item-by-store error on nine data sets, dropping LightGBM on eight and dropping TimesFM on seven; fitted weights lower the error most on frozen entrees, bottled juice and snacks. Fitted to the item-by-store error, the mean weights are 0.31 for the world model, 0.38 for LightGBM and 0.31 for TimesFM given the plan (Figure 9b); fitted to the total, TimesFM's weight falls to 0.24, and the weights move from data set to data set.

11 Occlusion: which plan inputs each model uses

Both tuned models forecast from the plan. Setting the price to the regular price makes the item-by-store error of the world model worse on eight of nine data sets and of LightGBM on eight (Favorita carries no price, so its forecasts do not change); setting every plan input to nothing planned makes it worse on nine and nine (Table 12). With the plan removed, the world model's forecast on the rows where the plan had something on falls by 7% to 73%, and LightGBM's by 4% to 73%: the two read about the same lift from a promotion. Cutting the

Table 10: Every model on the rolling windows: WRMSSE averaged over all levels, mean over windows. WM and LGB: the tuned world model and LightGBM; Avg: their average; Ens: the three-model ensemble with TimesFM given the plan; TFM+ and TFM: TimesFM with and without the plan; Chr: Chronos-Bolt; TST+ and TST: Timestral with and without the plan; TFT: the Temporal Fusion Transformer, with the settings of the first version; SN: seasonal naive. The comparison models are scored on cleaned prices on Dominick's; on M5, Frat and Favorita they are the runs of the first version on the same windows, since nothing in them was tuned. n/a marks a model not run on that data set. Shading marks the lowest: world model, LightGBM, average.

Data set	WM	LGB	Avg	Ens	TFM+	TFM	Chr	TST+	TST	TFT	SN
Analgesics	0.699	0.702	0.687	**0.676**	0.723	0.804	0.846	13.892	1.341	0.817	1.374
Bottled juice	0.575	0.581	0.551	**0.551**	0.673	0.791	0.802	3.889	1.012	0.741	1.279
Cheese	0.502	0.532	0.477	**0.471**	0.612	0.727	0.765	3.437	0.770	0.600	1.187
Frozen entrees	0.357	0.338	**0.332**	0.349	0.476	0.529	0.569	1.781	0.571	0.457	0.677
Soft drinks	0.705	0.680	**0.650**	0.655	0.770	0.909	0.976	2.281	0.858	0.815	1.001
Snacks	0.893	**0.778**	0.790	0.795	0.855	1.079	2.316	8.666	3.277	0.959	1.399
Frat	0.490	0.494	0.469	**0.454**	0.482	0.691	0.722	1.002	0.685	0.571	1.038
M5	0.709	0.697	**0.658**	0.729	0.982	0.964	1.184	1.038	0.988	n/a	1.146
Favorita	0.428	0.448	**0.403**	0.419	0.501	n/a	n/a	0.515	0.548	n/a	0.662

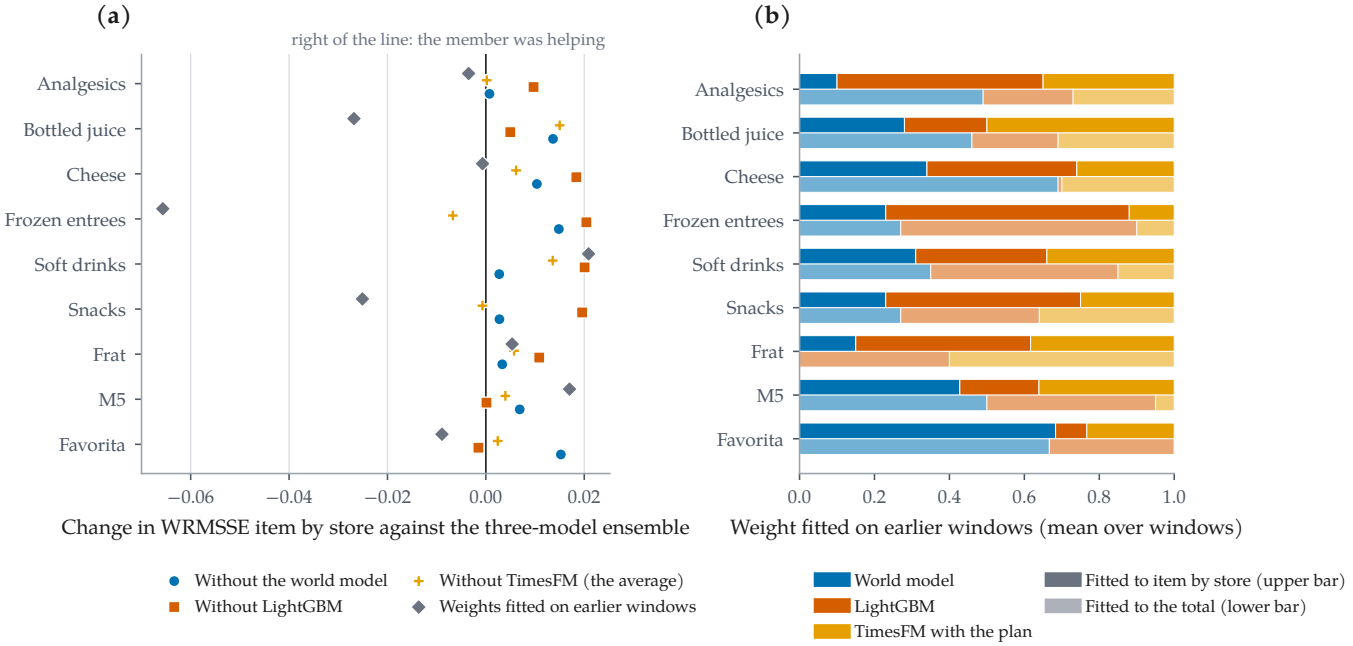


Figure 9: What each member adds to the three-model ensemble. (a) Change in WRMSSE item by store, mean over windows, when one member is dropped or the weights are fitted on earlier windows, against the ensemble with equal weights; right of the line, the dropped member was helping. (b) Weights fitted on earlier windows, mean over windows, when fitted to the item-by-store error (upper bar) and to the total (lower bar).

world model's history to the last few periods makes its error worse on nine of nine, so its carried state draws on more than the latest sales.

Figure 10 breaks the plan into its parts. On the Dominick's categories and Frat, price and the promotion flag carry the plan; display, feature advertising and a price cut on its own add less on Frat, the one data set that records them. The promotion flag helps the world model on eight of the eight data sets that carry it and LightGBM on seven. Events, SNAP days and paydays change the error by at most 0.019. Shuffling an input across the periods of each series, which keeps how often it is on but breaks its timing, moves the error the same way as switching it off for 43 of 46 pairs of input and data set. Keeping only the last few periods of history raises the world model's error on nine of nine data sets at the shortest cut (Figure 10c). The cut lengths are 4, 13 and 26 periods on every data set but M5, which used 28 and 56 days; on Favorita, which is daily, the shortest keeps four days and the error rises by 8.2.

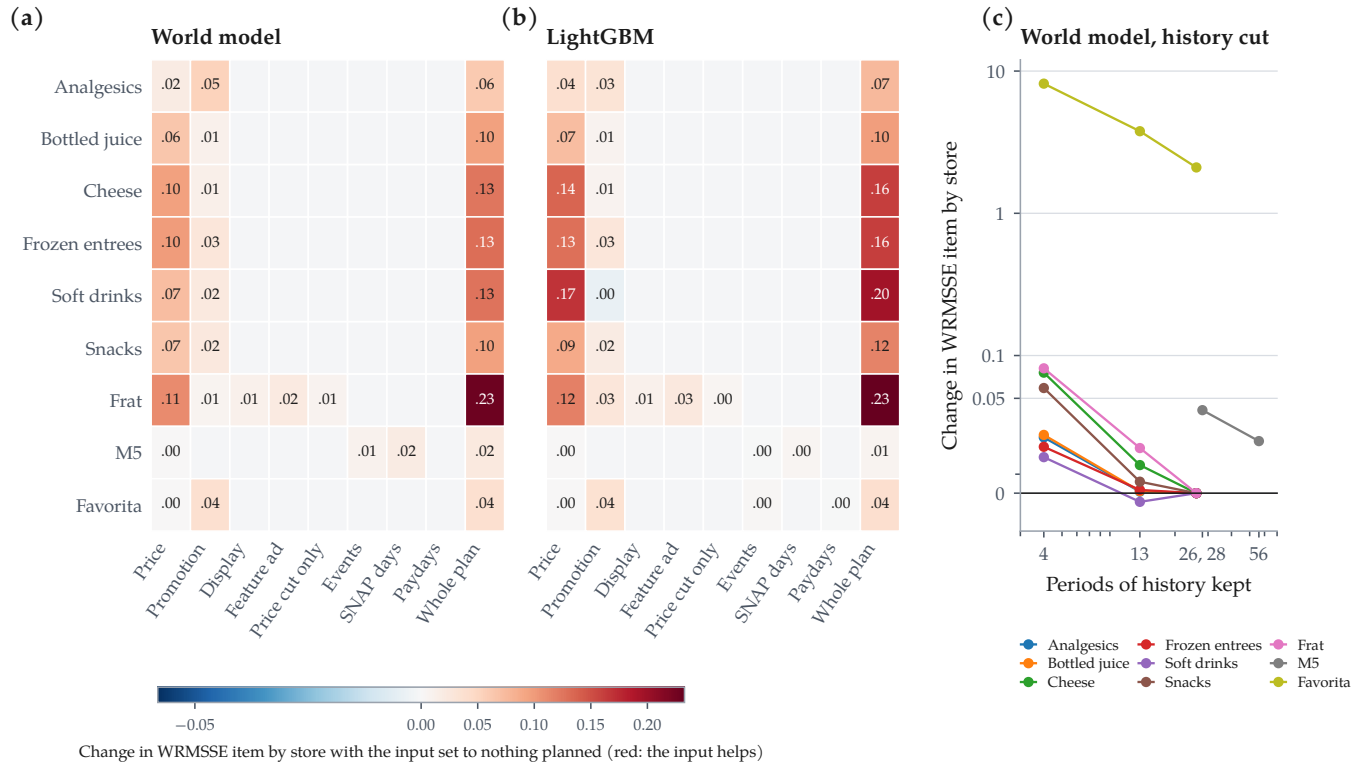


Figure 10: Which plan inputs the tuned models use. (a, b) Change in WRMSSE item by store, mean over windows, when one input is set to nothing planned (the price to the regular price), for the world model and LightGBM; grey cells are inputs a data set does not carry. (c) The world model's change in WRMSSE item by store when its history is cut to the last 4, 13 or 26 periods (M5: 28 or 56 days); the vertical scale is linear below 0.05 and logarithmic above.

Table 11: What each member adds to the three-model ensemble: WRMSSE item by store, mean over the rolling windows, of the ensemble, of the ensemble without each member (without TimesFM it is the average of the two tuned models), and of the ensemble with weights fitted on earlier windows. Lowest in bold.

Data set	Three	No WM	No LGB	No TFM	Fitted
Analgesics	0.655	0.655	0.664	0.655	**0.651**
Bottled juice	0.584	0.598	0.589	0.599	**0.558**
Cheese	0.599	0.609	0.617	0.605	**0.598**
Frozen entrees	0.401	0.416	0.422	0.395	**0.336**
Soft drinks	**0.547**	0.550	0.567	0.561	0.568
Snacks	0.556	0.559	0.576	0.556	**0.531**
Frat	**0.544**	0.547	0.555	0.550	0.549
M5	**0.832**	0.839	**0.832**	0.836	0.849
Favorita	0.687	0.702	0.685	0.689	**0.678**

12 Carried-state tests: what the world model's state does

The tuned world model uses the timing of the plan, and carries a promotion forward as more sales rather than a dip. Shuffling the plan's timing within each series makes its item-by-store error worse on all nine data sets, by 2% to 64% (Table 13): the forecast reads when a promotion runs, not only that one runs. A promotion and a price cut added in one period raise that period's forecast and still raise the next on six of nine data sets (the change in the next period runs from -4% to +54% of the forecast without them), where shoppers who stocked up would buy less; the dip after a promotion is not in the forecast, as in the first version. The state rolled through the window predicts sales after it only a little better than the state before it (an R^2 gain of 0.01 to 0.04).

13 Calibration and reliability of the forecast ranges

The ranges are calibrated item by store and too narrow above the store. Each model's ranges come from its own errors in the earlier windows (Section 6). Item by store, the 80% ranges of all three forecasts cover close to 80% of sales across data sets and windows (79%, 79% and 79% for the world model, LightGBM and the average), and the average has the lowest scaled pinball loss (0.230, against 0.241 and 0.235). At the chain total, where each window adds one error to learn from, the world model's ranges are too narrow (75% for 80%) and the average's are close to target (80%); the coverage of a single data set swings with so few earlier windows (Table 15).

Table 12: Which plan inputs each tuned model uses. Change in WRMSSE item by store, mean over windows, when the price is set to the regular price (price off) and when every plan input is set to nothing planned (plan off); the change in forecast units on the rows where the plan had something on; and, for the world model, the largest change when its history before the last few periods is removed. A positive change means the input was helping.

	World model			LightGBM			World model
Data set	Price off	Plan off	Units, plan rows	Price off	Plan off	Units, plan rows	History cut
Analgesics	+0.019	+0.061	-47%	+0.037	+0.075	-51%	+0.036
Bottled juice	+0.065	+0.095	-53%	+0.072	+0.098	-55%	+0.102
Cheese	+0.095	+0.125	-52%	+0.136	+0.161	-58%	+0.107
Frozen entrees	+0.100	+0.130	-73%	+0.129	+0.162	-72%	+0.044
Soft drinks	+0.073	+0.127	-61%	+0.168	+0.195	-73%	+0.085
Snacks	+0.065	+0.096	-37%	+0.089	+0.116	-44%	+0.126
Frat	+0.110	+0.229	-46%	+0.117	+0.233	-45%	+0.104
M5	+0.001	+0.023	-7%	+0.001	+0.006	-4%	+0.061
Favorita	+0.000	+0.037	-33%	+0.000	+0.040	-28%	+12.887

Table 13: The tuned world model's carried state, mean over windows. Timing: change in WRMSSE item by store when the plan's timing is shuffled within each series. Carry-over: forecast change in periods 1, 2 and 4 after a promotion and a 20% price cut added in period 1 to series with nothing planned, as a share of the forecast without them. Probe: R^2 of log sales after the window from the state before the window and from the state rolled through it (first of the five copies).

Data set	Timing	Period 1	Period 2	Period 4	Probe before	Probe after
Analgesics	+16%	+199%	+13%	+11%	0.80	0.81
Bottled juice	+29%	+188%	+6%	+9%	0.86	0.88
Cheese	+34%	+109%	-4%	+3%	0.91	0.92
Frozen entrees	+64%	+748%	+54%	+17%	0.80	0.84
Soft drinks	+46%	+327%	+34%	+10%	0.85	0.86
Snacks	+24%	+129%	+10%	+9%	0.86	0.87
Frat	+52%	+59%	-2%	-1%	0.86	0.89
M5	+2%	n/a	n/a	n/a	0.64	0.65
Favorita	+5%	+48%	+7%	+4%	0.75	0.77

Item by store, the 50% ranges hold 54%, 55% and 55% of sales for the world model, LightGBM and the average, and the 95% ranges 95%, 95% and 95%, so all three lie on the diagonal on average. Data set by data set the 80% coverage item by store runs from 71% to 94% for the average. At the total the world model's 80% coverage runs from 48% to 92% across data sets; a single window at the total holds one actual against one range, so its coverage is anywhere from 0% to 100%, and the 3 to 5 scored windows of each data set give a coarse estimate. By level (Figure 11f), the 80% ranges are too narrow at the store (65%, 68% and 66%) and for the item across stores (71%, 70% and 71%): the earlier windows' errors at those levels understate the next window's. A planner who orders from the 80% range at item by store gets what it says; one who plans store labour from the store-level range needs a wider one.

The whole distribution. Coverage checks three ranges; Figure 11 checks every quantile of the forecast distribution and scores it whole (Table 14). Item by store, 27% of actuals fall below the average's 10th percentile, 60% below its median and 82% below its 90th: the lower quantiles of a slow item are zero and many actuals are zero too, so the curve is read at the midpoint of the shares strictly below and at or below each quantile. At the total, 11% of actuals fall below the world model's 10th percentile and 86% below its 90th; for LightGBM the shares are 7% and 88%. The ranges at the total lie close to the diagonal with the upper tail a little short; item by store the upper tail is shorter, with 82% of actuals below the 90th percentile. On the continuous ranked probability score (CRPS), which scores the whole distribution, the average is lowest item by store on nine of nine data sets, LightGBM on none and the world model on none; at the total the average is lowest on five, LightGBM on three and the world model on one.

The chance of selling nothing. The ranges do not forecast whether an item sells at all. Built from errors scaled to each series, they put the chance of a zero too low at every level: in each group of forecast chance, more node-periods sold nothing than the chance says (Figure 11d). Their Brier score is 0.190 for the average, 0.190 for the world model and 0.190 for LightGBM, against 0.138 for each series' own share of zero-sale periods over the year before; the models beat that share on Frat only (Table 14). A retailer who needs the chance of no sale, to check availability or prune the range, should forecast it directly: the hurdle likelihood kept on M5 and Favorita (Section 15.1) does so inside the world model, but these ranges use only its point forecast.

The reliability of a single run is a separate question: the world model's scores vary from seed to seed, by a stan-

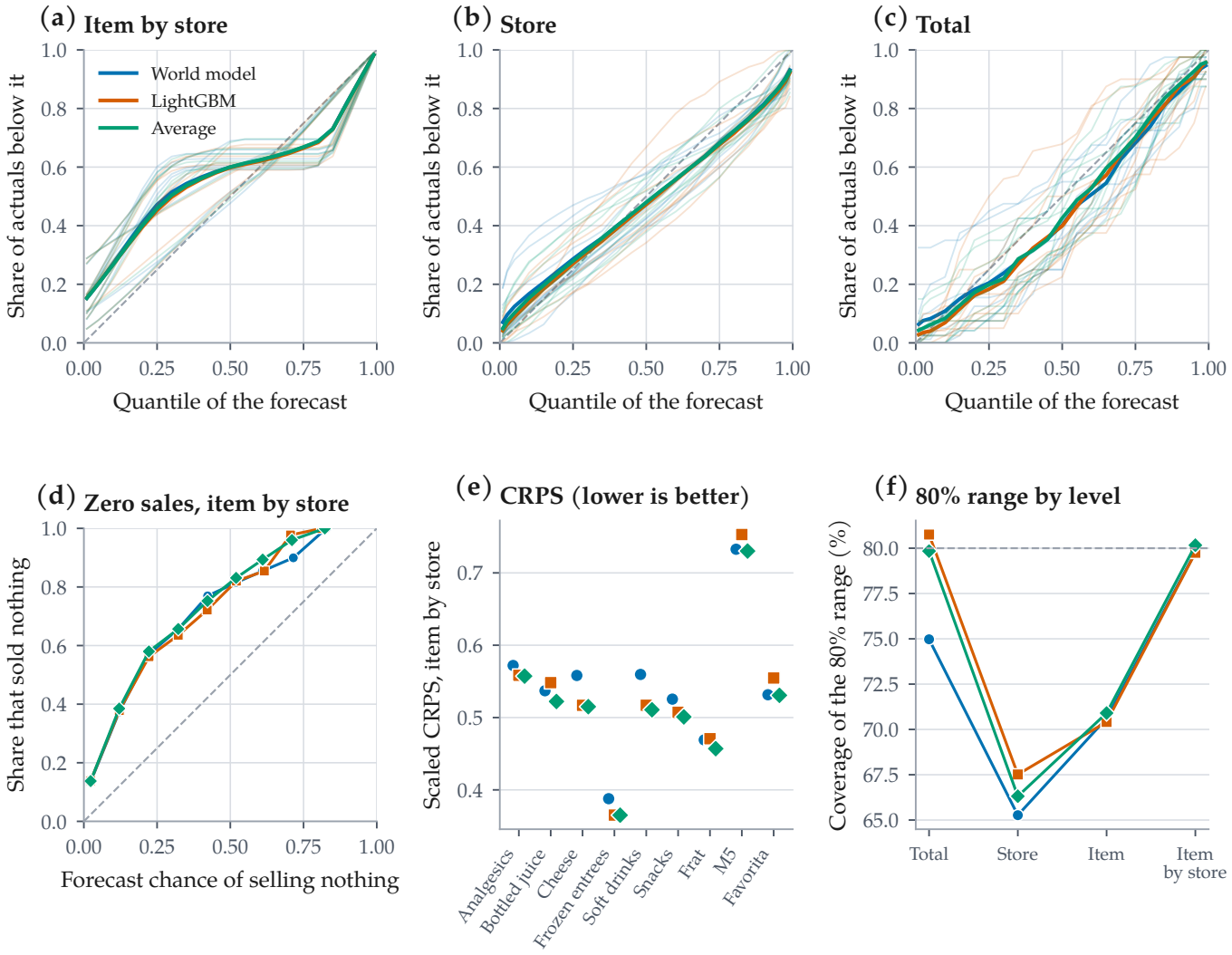


Figure 11: Reliability of the whole forecast distribution. (a to c) Share of actuals below each quantile of the forecast, item by store, at the store and at the total, for the world model, LightGBM and the average; faint lines are data sets, solid lines their mean, and the dashed line is perfect calibration. For whole-unit sales each share is the midpoint of the shares strictly below and at or below the quantile. (d) Reliability of the forecast chance of selling nothing, item by store: node-periods grouped by their forecast chance in tenths, pooled over data sets and windows (groups of fewer than 100 left out). (e) Scaled CRPS item by store, mean over windows. (f) Coverage of the 80% range by level, mean over the nine data sets; the dashed line is the stated 80%.

dard deviation of 6.3% of the score on the Dominick's categories for a single fit and 1.8% for the five-copy average the tuned model uses (Figure 7). Every comparison in this paper uses the five-copy average and three seeds where a change is tried.

14 Decision costs and reconciliation

Decision costs. The average gives the lowest category-plan cost on seven of nine data sets and LightGBM on the other two (frozen entrees and snacks); the world model alone is lowest on none (Table 16). Ordering each forecast's 0.8 quantile for every item and store, with each unit over costing 1 and each unit short 4 (Section 6), gives a lower simulated loss with the average than with LightGBM on six data sets and less than with the world model on seven. Both are simulations under stated costs, not business results: the ordering loss is for a single period, with no stock carried over, no lead time and no shelf life, and the category-plan cost is a forecast error in dollars, not a realised commercial outcome. They rank the forecasts by a stated cost; they do not establish inventory savings in a deployed retailer, which would need a multi-period inventory system with the retailer's own stock records.

Reconciliation. Summing the average upward is more accurate over all levels than the best reconciled ETS forecast on seven of nine data sets (Table 17). The exceptions are soft drinks and snacks, where reconciled ETS is more accurate at the coarse levels, which carry most

Table 14: Probabilistic scores of the forecast ranges, mean over windows. Scaled CRPS item by store and at the total, from each model's point forecast and the quantiles of its scaled errors in earlier windows on a grid of 23 quantiles, weighted by revenue as WRMSSE. Brier score of the forecast chance of selling nothing, item by store, against each series' share of zero-sale periods over the year before (Past). Lowest in bold.

	CRPS, item by store			CRPS, total			Brier score, zero sales				Periods
Data set	WM	LGB	Avg	WM	LGB	Avg	WM	LGB	Avg	Past	with none
Analgesics	0.572	0.558	**0.557**	0.622	0.598	**0.591**	0.395	0.388	0.393	**0.199**	52%
Bottled juice	0.537	0.548	**0.522**	0.397	0.354	**0.335**	0.129	0.132	0.131	**0.124**	17%
Cheese	0.558	0.517	**0.515**	0.270	0.340	**0.264**	0.110	0.109	0.110	**0.090**	13%
Frozen entrees	0.388	0.365	**0.365**	0.249	**0.208**	0.209	0.205	0.204	0.205	**0.165**	30%
Soft drinks	0.560	0.517	**0.511**	0.630	0.626	**0.564**	0.185	0.192	0.188	**0.151**	30%
Snacks	0.525	0.507	**0.501**	1.430	**0.898**	1.047	0.194	0.189	0.192	**0.148**	26%
Frat	0.469	0.471	**0.457**	**0.248**	0.344	0.275	0.023	0.024	**0.023**	0.051	8%
M5	0.733	0.753	**0.730**	0.427	**0.336**	0.360	0.334	0.326	0.331	**0.195**	56%
Favorita	0.532	0.555	**0.531**	0.163	0.160	**0.128**	0.137	0.143	0.140	**0.117**	20%

Table 15: Coverage (%) of each model's 80% interval, from its own errors in earlier windows, item by store and at the total, mean over the scored windows.

	Item by store			Total		
Data set	WM	LGB	Avg	WM	LGB	Avg
Analgesics	91	91	91	82	82	82
Bottled juice	72	73	73	85	98	92
Cheese	74	74	75	75	95	92
Frozen entrees	70	71	71	88	82	90
Soft drinks	75	75	76	68	68	68
Snacks	73	73	74	48	75	65
Frat	88	87	88	92	62	75
M5	94	94	94	73	80	73
Favorita	81	80	80	65	85	81

Table 16: Decision costs of the tuned models. Store orders: cost of ordering each forecast's 0.8 quantile, summed over windows, relative to LightGBM's. Category plan: absolute error in dollars over sales in dollars at category by store (%), mean over windows; lowest shaded.

	Store orders, ratio 0.8			Category plan (%)		
Data set	WM	LGB	Avg	WM	LGB	Avg
Analgesics	1.012	1.000	0.996	10.7	11.0	**10.6**
Bottled juice	0.983	1.000	0.965	13.9	14.0	**13.2**
Cheese	1.077	1.000	1.006	10.5	12.1	**10.2**
Frozen entrees	1.022	1.000	0.985	20.8	**18.6**	18.8
Soft drinks	1.070	1.000	1.004	20.7	21.7	**19.5**
Snacks	1.040	1.000	1.000	26.5	**19.7**	21.7
Frat	1.015	1.000	0.981	15.3	15.0	**14.5**
M5	0.972	1.000	0.977	10.1	10.0	**9.4**
Favorita	0.952	1.000	0.960	9.8	9.7	**8.7**

of the all-level score; combining either tuned model with ETS above it does not do better than the average on most data sets. The comparison is with reconciled ETS only, which does not read the price and promotion plan; it shows that summing the average upward is a reasonable default here, not that it beats reconciling the tuned models themselves, which was not tested.

15 What was tried and dropped

Most of what was tried did not stay. The searches of Section 7 kept a small share of the changes they ran (Table 5); every change tried alone in the first round is in Figure 5, and the losses tried in the first version are in Table 4. Table 18 lists the larger things the revision tried beyond the searches and did not keep, and when each becomes worth trying again.

The first version of this paper also tried a stock term for the dip after a promotion, extra loss weight on the periods after a promotion, and a graph model sharing information between stores; none was kept, and they were not run again on the tuned models. Their record is in the first version; the loss experiments among them are in Figure 12.

15.1 Losses and pretraining explored

Count likelihoods helped on three data sets and JEPA pretraining on one; the plain loss stayed everywhere else. After each search, the world model had a round of count likelihoods and a round of JEPA pretraining (Section 7). The likelihood round screened four likelihoods on each of nine data sets with one seed each, 36 screens in all, of which 11 came in below the starting score; the best went on to three seeds, and a change was kept only where it beat the start by more than the keep bar (Fig-

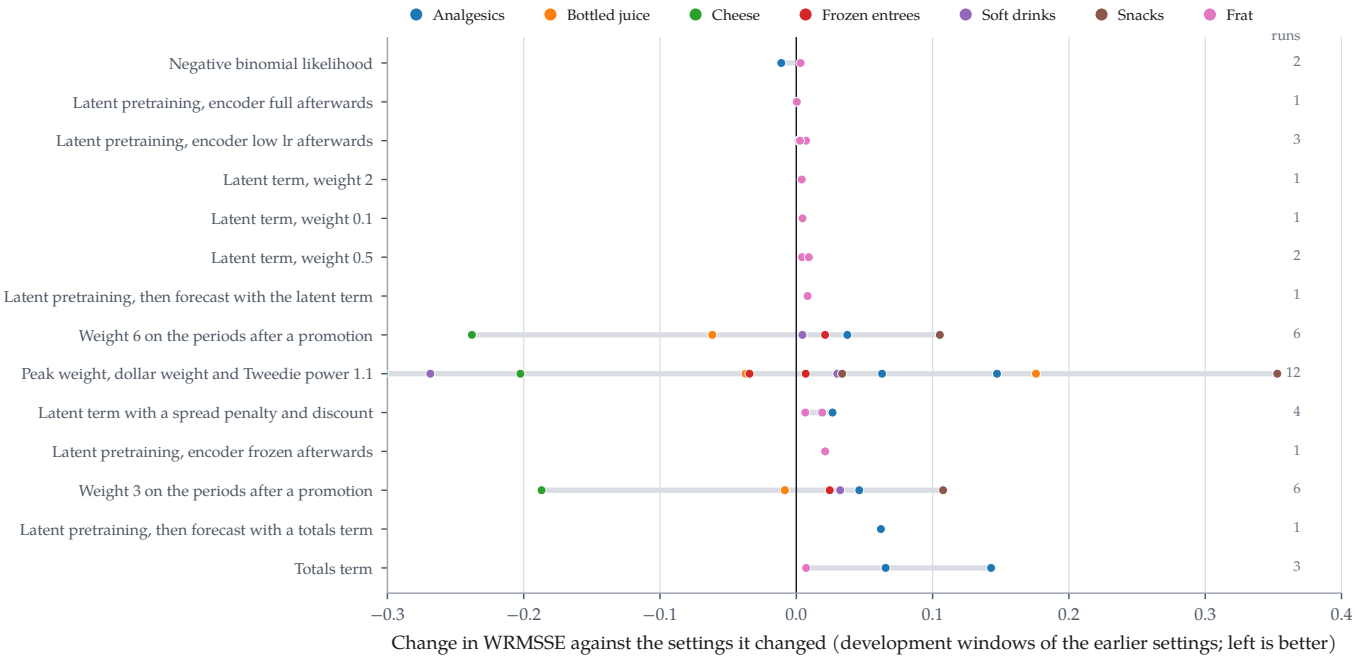


Figure 12: Every loss change of the first version against the settings it changed, on the development windows. Each dot is one run, coloured by data set; the bar spans the runs of a change, and the right-hand column counts them. Changes not run on the full data are not shown.

Table 17: Reconciliation against summing one forecast: WRMSSE averaged over all levels, mean over windows, of the best reconciled ETS forecast, of each tuned model combined with ETS above it by weighted least squares, and of the average summed upward.

Data set	Best ETS	WM + ETS	LGB + ETS	Avg
Analgesics	0.836	0.903	0.933	**0.687**
Bottled juice	0.878	0.849	0.853	**0.551**
Cheese	0.742	1.430	1.215	**0.477**
Frozen entrees	0.549	0.510	0.474	**0.332**
Soft drinks	**0.617**	1.061	0.907	0.650
Snacks	**0.698**	0.979	0.805	0.790
Frat	0.693	0.499	**0.466**	0.469
M5	0.945	0.821	0.813	**0.658**
Favorita	0.570	0.434	0.462	**0.403**

ure 13a and Table 19). It was kept on three: Snacks, M5, Favorita. The JEPA round pretrained the encoder on the latent term alone (Section 5.2), for a short and a long run, then fitted the forecast at a lower rate. Of its 18 screens, 2 came in below the start, and it was kept on Snacks only (Figure 13b).

The first version tried 16 changes to the world model's loss in 60 runs on the development windows, 44 of them scored against the settings they changed (Figure 12); the other 13 were not run on the full data. 10 of the 44 lowered the error. Latent pretraining and the latent term, in every form tried, moved the score by at most 0.062 either way. Extra weight on the periods after a promotion and the combined peak, dollar and Tweedie weighting moved it most, in both directions: every form helped cheese and hurt snacks and analgesics, so none was kept for every category. The settings searches of Section 7 then tried the loss weights again one at a time on each data set (Figure 5).

16 Discussion

Table 20 maps the rule of Section 8.6 to the decisions a retailer makes from the forecast. Each row is what the tuned models did on the nine data sets studied, within the scope of Section 17.

Why the average works. The two tuned models are close in accuracy and wrong in different places. Across the nine data sets, each is clearly more accurate than the other in a similar number of paired comparisons (57 for the world model, 63 for LightGBM), and the data sets on which each leads differ. An equal-weight average of two forecasts whose errors differ is more accurate than either on most series, and no weights have to be estimated, which the forecast-combination literature finds is hard to beat [72, 73].

What the world model brings. The world model's contribution is clearest where the forecast has least to go on. On items it was not trained on it lost at most 0.040 in scaled error, against up to 0.311 for LightGBM, and forecast within 11% of actual sales; LightGBM forecast as lit-

Table 18: What the revision tried beyond the searches and did not keep. Each was written down before it ran.

Tried	What happened	Worth revisiting when
Count likelihoods that treat zero weeks on their own (negative binomial, zero-inflated, hurdle), on the world model	Kept on M5 and Favorita (hurdle) and snacks (negative binomial); nothing on the other six data sets	Lines are slower or more often out of stock than here
Pretraining the world model's encoder on the latent objective, then fitting the forecast at a lower rate	Kept on snacks only, on the search windows; with nothing kept on eight of nine data sets, Favorita's rerun on its final loss was stopped early	More unlabelled history is available than the forecast needs, as across many categories
Training on other retailers' data with each own data set (RQ4)	Helped neither model by more than 0.003 on any data set, and hurt LightGBM most	The other data come from the same retailer or the same categories
The world model alone for series with less than 26 weeks of history	No better than the average on lines with history; worse on snacks	A longer history threshold is set from a retailer's own range data
Weights fitted to earlier windows in place of equal weights	Reported in Section 10	Many more windows are available to fit them on
Timestral in place of TimesFM in the three-model ensemble	Reported in Section 10	A retail-trained version is released

Table 19: Every likelihood and JEPA pretraining recipe tried on the tuned world model, WRMSSE on the search windows. Start: the settings each round began from. A number in brackets is a one-seed screen that did not earn a three-seed try; a plain number is the three-seed mean; bold marks the change kept, which had to beat the start by more than the keep bar on the search windows.

	Likelihood round					JEPA pretraining round		
Data set	Start	NB	NB, spread by item	ZINB, spread by item	Hurdle NB, spread by item	Start	Short	Long
Analgesics	0.509	(0.518)	(0.515)	(0.518)	(0.539)	0.509	0.511	(0.537)
Bottled juice	0.587	(0.592)	(0.601)	(0.614)	(0.604)	0.587	(0.635)	(0.615)
Cheese	0.392	(0.450)	0.403	0.396	0.382	0.392	(0.401)	(0.406)
Frozen entrees	0.216	0.214	0.218	0.215	0.216	0.216	(0.267)	(0.255)
Soft drinks	0.572	(0.613)	(0.604)	(0.606)	(0.598)	0.572	0.612	(0.603)
Snacks	0.439	(0.481)	**0.420**	0.427	0.429	0.420	(0.428)	**0.400**
Frat	0.547	(0.584)	(0.600)	(0.585)	(0.590)	0.547	(0.572)	(0.566)
M5	0.547	(0.553)	(0.558)	0.552	**0.533**	0.533	(0.537)	(0.544)
Favorita	0.483	0.488	(0.476)	0.469	**0.468**	0.483	(0.527)	(0.523)

tle as two thirds. It reads an item's recent sales into a state and carries that state forward under the plan, so it does not need the item in its training data, whereas LightGBM takes the item as one of its labels and has learned nothing for an item it was not trained on. The same carried state keeps its totals over the promotion block closer to actual sales than LightGBM's on M5, Favorita and soft drinks.

What the trees bring. LightGBM is the more accurate of the two on frozen entrees and snacks at every grain, and its block totals are closer than the world model's on every Dominick's category but soft drinks. It needs no pretraining and trains in minutes, and on lines with history it is as accurate as the world model on most data sets. On snacks, where the world model alone is weakest (0.893 overall against 0.778 for LightGBM), the average stays close to LightGBM at 0.790.

Why run both. Running both models means a second pipeline, a GPU-trained network beside trees that train in minutes, and on lines with history the average gains little over the better single model. The case for it rests on totals: over the promotion block LightGBM alone drifts in total on M5, Frat, Favorita and soft drinks, where the world model does not. On M5 its block error is 0.953, against 0.598 for the world model and 0.589 for the average (Section 8.1). A retailer whose plans are summed to the category and the chain pays for that drift at every level above the item. The searches behind the tuned settings took 4,305 hours of run time across the nine data sets (Section 19), a one-off cost. In the one transfer between retailers tested, settings tuned on M5 carried to Favorita (Section 8.4), so a retailer may not need to repeat the search; one transfer does not establish that, and a retailer should check it on its own data.

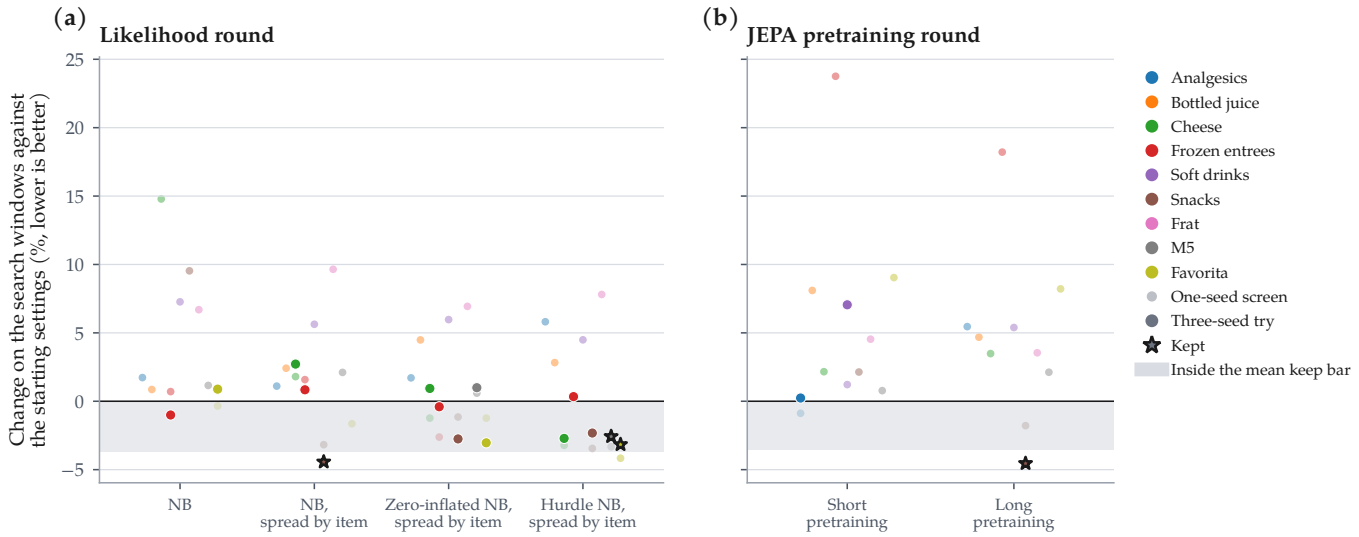


Figure 13: The likelihood and JEPA pretraining rounds of the tuned world model. Change in WRMSSE on the search windows against the settings each round started from, for each one-seed screen (faint) and three-seed try (solid); stars mark the change kept. A change was kept only where its three-seed mean beat the start by more than the keep bar; the grey band is the mean keep bar over data sets.

Table 20: The decisions a retailer makes from one item-by-store forecast, and what the tuned models did at each on the nine data sets studied.

Decision	Grain	Forecast to use	What the tuned models did
Store orders and inventory	Item by store, by day or week	The average	Lowest error at the item levels on all nine data sets; clearly less accurate than either model in at most 8 of 100 paired comparisons; simulated order cost below LightGBM's on six of nine
Promotion and supplier funding	Item across stores, by week and over the promotion block	The average	Lowest at the item levels on all nine; over the whole block lowest on five and second on the other four; LightGBM alone drifts in total on Frat, M5, Favorita and soft drinks
Category performance	Category by store and category, by week	The average	Lowest at store grain and above on eight of nine; LightGBM lowest on snacks; lowest category-plan cost on seven of nine
Financial plan	Chain total	The average	As for category performance; the total is the sum of the same item-by-store forecast, so it agrees with every plan below it
New items and range changes	Item by store, items not yet trained on	The world model alone	Clearly more accurate than the average on all five data sets tested; LightGBM forecast 66 to 93% of actual sales on these items
New stores	Item by store, stores not yet trained on	The average	The average and the world model each clearly ahead on one data set and level on three; LightGBM clearly behind the world model on four of five
Choosing for another retailer	Any	Rerun the comparison	The order of models is a finding on these data; every comparison, including those that lost here, should be rerun on the retailer's own data

17 Limitations and threats to validity

Scope: two of the four Ps, on public data. The paper addresses price and promotion. It forecasts items newly added to a range but does not decide product (which items to range) or place (space, distribution), and it does not solve the four Ps together; a unified forecast across them does not exist in this work, and the study does not show which model would work best at each grain under one. The nine data sets come from four retailers, and six of them are categories of one retailer (Dominick's), so the evidence across retailers rests on four. The ordering of models is not established for a single retailer's data, with its own levers, stock and space records, and the grain comparison should be repeated on it before a choice is made.

Adapting the findings. The rule of Section 8.6 and the ordering of models behind it hold on the nine data sets studied. They are not a ranking of methods for retail in general. A retailer or researcher adopting the framework should rerun every comparison on its own data, including the models and combinations that lost here, since data with other levers, lines and lengths of history can change the order. The code and settings for every comparison are released for that purpose.

Forecast horizon. Every forecast in the study runs 8 weeks ahead on the weekly data and 28 days ahead on the daily data. Whether the models' accuracy, and the order between them, holds over the 26- and 52-week horizons of range, space and supplier reviews was not tested, nor whether accuracy falls with the horizon at the same rate for each model. Testing it needs a retailer's own plans and range decisions to score against and is left for separate work.

New items. The kept-out items are existing items held out of training, not new products. Each model read their recent sales before forecasting them. An item with no sales at all, on its first day in range, was not tested; the new-item part of the rule covers items the models have not been trained on, not new products with no history; a new product launch, with no sales and often no like item, needs its own test. That part of the rule was drawn from the results, after a wider rule written down in advance held only in part (Section 8.6).

Windows and seeds. Four to six windows per data set is a thin basis for a bootstrap over windows and for a model confidence set: the intervals are approximate, the set has little power, and a model kept in it has not been shown to be as good as the best. Adjacent windows share training history and season, so they carry less information than their number suggests; the item bootstrap within a window treats the items as the sample and does not cover variation between windows. Averaging five fits with different seeds removes much of the noise between runs that the searches measured (Section 7.4), but both models on M5 are single fits.

Analyses added after the first results. The analyses by line type, the whole-block view, the forecast by history and the supporting reruns were added after the first scores of the tuned models were read. Each was written down before it ran, but which to run was chosen knowing the first results, and the new-item part of the recommended rule was drawn from the results (Section 8.6).

Comparison models. Only the world model and LightGBM were tuned; the Temporal Fusion Transformer runs with the settings of the first version, and the pretrained models as published. The comparison between the two tuned models is therefore the fair one, and the others are reference points. TFT was not run on M5 or Favorita, and Chronos-Bolt and TimesFM from history alone not on Favorita. TimesFM with the plan fits its linear plan model within groups of 2,048 series. The pretrained models' point forecast was fixed in advance as the mean of their output, which for Timestral let a few extreme upper quantiles dominate on weekly data.

What lift means. Merchants promote in weeks they expect to sell well, so part of a measured lift is good timing. The lift is what the forecast attributes to the plan, not what the plan caused. Our companion audit [71] measures the gap in generated store data where the true lifts are known. Plan responses in this paper, including the perturbation tests, are forecasts, not causal estimates.

What the data do not record. None of the data sets records stock, so a sell-out looks like low demand; or shelf space. Favorita records no prices. The data were collected between 1989 and 2017 and their plan inputs differ, so the results compare models within each data set. The decision costs are single-period simulations under stated costs, not a dynamic inventory system, and they do not establish savings in a deployed retailer.

18 Conclusion

A retailer makes decisions from the same demand at different grains, and on these data no single model was the most accurate at every grain. One forecast made item by store and summed upward, as the equal-weight average of a tuned world model and tuned gradient-boosted trees, was the most accurate choice at store grain and above on eight of nine data sets and at the item levels on all nine, on the fast lines and on the slow tail. The combination held across grains, but the choice of model changed for items absent from training: there the world model alone was more accurate than the average on every data set tested, because LightGBM with the item label under-forecast them. In the one transfer between retailers tested, settings tuned on M5 carried to Favorita, and training on other retailers' data did not help. The rule that follows, the average everywhere and the world model alone for items not trained on, is a finding on these data for forecasts of 8 weeks and 28 days under price and promotion plans. The comparison behind it, run over several windows with paired tests and every losing model reported, is what we propose a retailer reruns on its own data, and over its own planning horizons, before adopting it.

19 Reproducibility and compute

Records. All code, configuration files and tests are in the repository. A run refuses to start on uncommitted code and writes a manifest holding the code commit, the full configuration and its checksum, the seed, checksums of every data file read, package versions, the machine and run times. The scores and manifests of the final, transfer and pooled runs, every try of the searches, and the analyses of the tuned models are committed under `results/final/`, `results/ftransfer/`, `results/pooled/`, `results/research/` and `results/analysis_tuned/`; the plan and every dated entry are in `docs/PAPER1_REVISION_PLAN.md`. `make` in `paper/` rebuilds every figure, table and quoted number from these files (27 seconds from a fresh clone of the repository, 13 November 2026). Forecast files and model checkpoints are kept outside the repository on the results volume, with their checksums recorded.

Compute. The first version ran on one Google Colab machine with an NVIDIA Tesla T4 (Table 21: 94 timed runs, 44.7 machine-hours). The revision ran on rented machines: the searches, final runs, transfer tests and pooled training on GPUs from Vast, managed and analysed on Modal. Table 22 gives the run time recorded in each of the 4,870 run folders, 4,679 hours in all, most on RTX 4090 GPUs; because a resumed run records only its last stretch, this is a floor. The machine time paid for on Vast was 4,797 GPU-hours (US$2,229), and the revision cost US$5,342 in all from 26 October to 12 November 2026: US$2,862 on Vast, including storage and data transfer, and US$2,480 on Modal. The total includes every run made, the searches' screens, repeats after machine failures and runs lost when a machine stopped, not only the runs reported.

Table 21: Run times of the final and comparison runs, from their manifests. Wall-clock minutes per run (data loading, training, forecasting and scoring) and hours in total, on one Google Colab machine with an NVIDIA Tesla T4 (LightGBM uses its CPU). Dominick's: six categories together; a Dominick's world-model run averages three fits. Two M5 runs resumed after the Colab runtime dropped record only the resumed part (the shortest values in their rows). Not counted: runs saved before run times were recorded, the development study, and TimesFM on Dominick's and Frat.

Data	Model	Runs	Median	Range	Hours
M5	World model	9	101.5	37.3 to 103.2	14.0
M5	Lift model	9	88.7	10.3 to 94.6	12.2
M5	LightGBM	9	78.1	76.5 to 79.9	11.7
M5	Graph model	12	7.2	7.2 to 7.5	1.5
M5	TimesFM	8	3.9	3.8 to 4.1	0.5
M5	Seasonal naive	2	1.3	1.2 to 1.3	0.0
Dominick's	World model	18	9.2	9.0 to 10.6	2.8
Dominick's	LightGBM	18	4.6	3.1 to 11.5	1.7
Frat	World model	3	2.2	2.2 to 2.2	0.1
Frat	Lift model	3	2.0	1.9 to 2.0	0.1
Frat	LightGBM	3	1.5	1.5 to 1.6	0.1
All 94 timed runs					44.7

Table 22: Run time of the revision, from each run's manifest. Runs counts every run folder, the pieces of split runs included; hours is their summed wall-clock time; machines are the GPUs or CPU hosts the runs used, with the number of runs on each. A run resumed from a partway save records only its last stretch, so the hours are a floor; the machine time paid for is in the text.

Study	Runs	Hours	Machines
Searches of both models' settings	3,958	4,305	RTX 4090 (1182), RTX 5090 (54), L4 (868), CPU (1288), other GPUs (566)
Final runs on the rolling windows	245	41	RTX 4090 (220), RTX 5090 (16), other GPUs (9)
Transfer and new-retailer tests	358	150	RTX 4090 (252), RTX 5090 (85), other GPUs (21)
Training on other retailers' data	249	176	RTX 4090 (216), RTX 5090 (17), other GPUs (16)
Comparison models on cleaned prices	60	8	RTX 4090 (48), other GPUs (12)
All	4,870	4,679	

20 Data, code and licence

Data. M5 is available from the competition's Kaggle page [1]; Dominick's from the Kilts Center for Marketing, University of Chicago Booth School of Business [23]; Breakfast at the Frat from dunnhumby [24]; Corporación Favorita from its Kaggle competition [75]. Each is obtained from its provider under the provider's own terms. No data is redistributed with the code.

Code. The code is at `https://github.com/karthik-iyer-au7/retail-world-models`; the version used for this paper is archived as release v1.0.0 on Zenodo, doi:10.5281/zenodo.23448056.

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prohibited unless approved in writing by the authors.

21 Author contributions

Karthik Iyer conceived and scoped the work, designed the study, wrote the code, ran the experiments and analyses, and wrote the paper, with AI assistance as declared in §23. Madan Dabbeeru widened the range of methods considered for each research question, and reviewed the work and the manuscript.

22 Practice background and independence

The first author has worked in retail for fifteen years, across continents, in grocery, department stores, fashion, electronics and online marketplaces. No data, method, code or other intellectual property from those roles or engagements is used in this work, and no employer or client data are used: the study is built from the ground up on public data sets. What the authors carry from practice is the set of questions retailers ask of a forecast, which shaped the research questions in Section 2. The first author's research focus is world models; the second author widened the scientific options considered for answering each question.

23 Declaration of generative AI and AI-assisted technologies

The research questions, scope, study design, choice of data and methods, and interpretation of results are the authors' own. Claude (Anthropic) was used as an assistant to write and test code, run analyses, produce figures and tables, and draft and edit text. The authors directed this work throughout, reviewed every result, and corrected and revised the code, analyses and text over many iterations, and take full responsibility for the content of the publication.

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A Constraints the models work within

Table 23 lists the constraints. Each model's tuned settings are in Table 3.

Table 23: Every constraint the trained models and the ensemble work within, with its value and where it lives in the code.

Constraint	Value	Where
Log forecast bound (world model)	$[-10, 10]$ around the item's level	`src/rwm/model/state_model.py` readout
Lift when nothing is planned	Exactly zero	Readout; tested in the code's tests
Baseline under a changed plan	Unchanged	Readout; tested in the code's tests
Regular price	Highest price in the last 12 periods	Plan features
Brand-switching share α	One per run, fitted on the last 8 training weeks	Dominick's world model
Forecasts	Not below zero	Every model
Average weights	One half each	`scripts/revision_analysis.py` (`ensemble_2`)
Three-model ensemble weights	One third each, fixed 13 October 2026	`scripts/revision_analysis.py`
Forecast by history	World model alone under 26 weeks (182 days) of sales	`scripts/revision_analysis.py` (`by_history`)
Seed copies	Five fits averaged (one on M5)	`src/rwm/seed_average.py`
Interval pooling	Horizons pooled below 50 earlier errors per horizon	`src/rwm/evaluation/compare.py`
Interval ends	Rounded outwards to whole units (not Favorita)	`src/rwm/evaluation/compare.py`
TimesFM covariate fit	Groups of 2,048 series	`src/rwm/prior_work/`
Timestral context and batch	512 periods; batch 256, halved on running out of GPU memory	`src/rwm/prior_work/timestral_zero_shot.py`